

Required Report: Required - Public Distribution

Date: July 17, 2026

Report Number: E42026-0052

Report Name: Biofuels Annual

Country: European Union

Post: The Hague

Report Category: Biofuels

Prepared By: Bob Flach, Katarina von Witzke, and Sophie Bolla

Approved By: Laura Geller

Report Highlights:

EU bioethanol and Biomass-Based Diesel (BBD) consumption is rising due to stricter renewable transport targets enforced by the revised second Renewable Energy Directive (REDII). Though electrification, biofuel blending limits, crop caps, and specific sustainability requirements restrain growth. While domestic bioethanol production remains stagnant, BBD manufacturing is expanding, supported by anti-dumping duties on Chinese imports and the EU Sustainable Aviation Fuel (SAF) targets. Driven by growing demand and flat local output, EU bioethanol imports are forecast to increase in 2026, primarily from the United States and Ukraine. In 2026, EU BBD imports from China and Argentina are collapsing while volumes from India and Thailand surge. Following changes to American tax credits, the United Kingdom has emerged as the largest export market for EU BBD. Finally, the consumption of advanced biofuels derived from waste streams must increase significantly by 2030 to meet EU regulatory mandates.

Table of Contents

I. Summary	2
II. Policy and Programs.....	4
The EU’s Renewable Energy Directive (RED)	4
The Renewable Energy Directive II (REDII)	4
Sustainable Aviation Fuels (SAF)	9
Sustainable Maritime Fuels (SMF)	10
EU Climate Law	10
EU Deforestation-free Supply Chain Regulation (EUDR)	11
The EU Taxonomy for Sustainable Activities	11
Policy Response to the War in Ukraine.....	11
Market Access.....	12
III. Ethanol.....	14
EU Production, Supply and Demand Table	14
Consumption	16
Production and Capacity.....	19
Feedstock Use and Production of Co-products	20
Trade	21
IV. Biomass-based Diesel.....	22
EU Production, Supply and Demand Table	22
Consumption	24
Production and Capacity.....	25
Feedstock Use and Co-products Production.....	27
Trade	28
V. Advanced Biofuels.....	30
Hydrogenation-Derived Renewable Diesel (HDRD).....	31
Advanced Bioethanol	31
Sustainable Aviation Fuels (SAF) and Marine Fuels (SMF)	32
Commercial Production of Advanced Biofuels	33

VI. Notes on Statistical Data..... 37

 Bioethanol..... 37

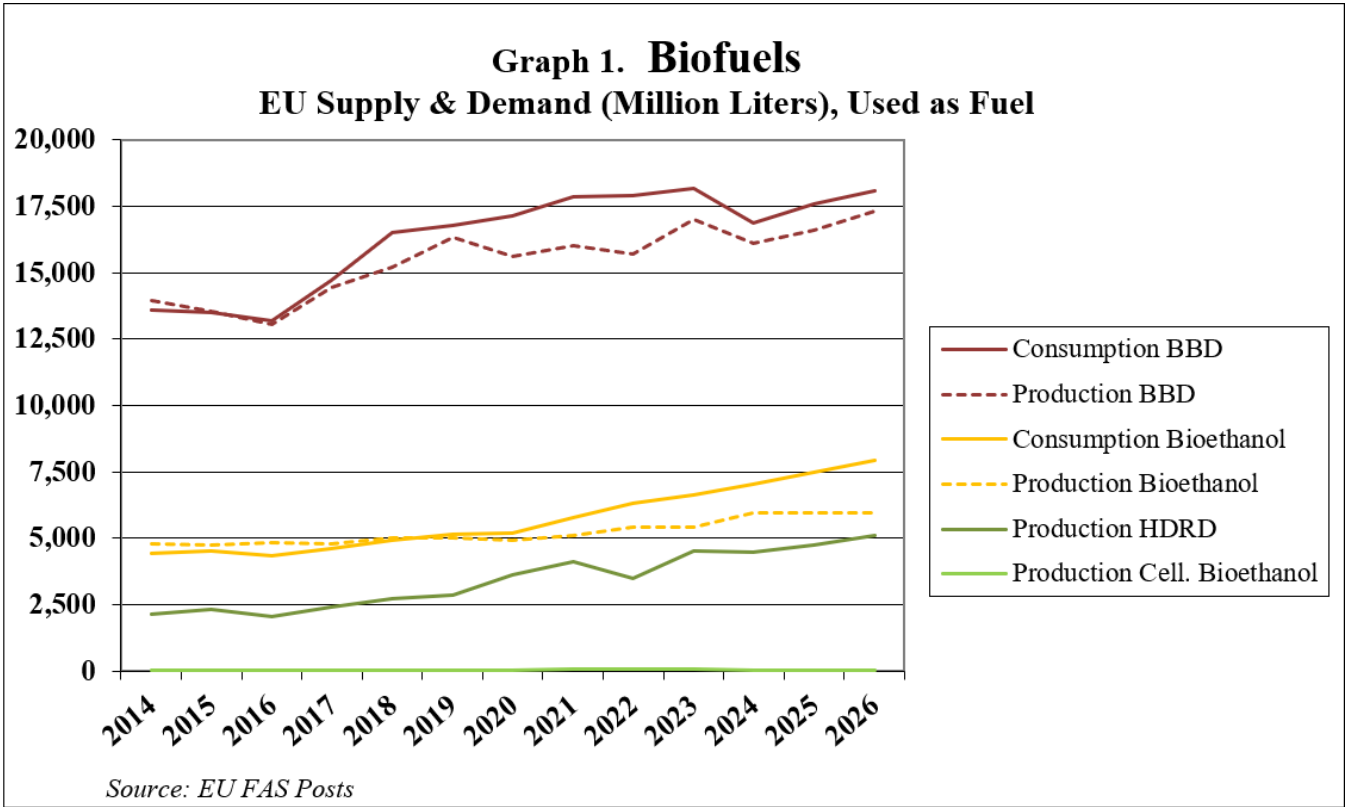
 Biodiesel/HDRD 38

Appendix I – Abbreviations..... 38

Appendix II - Energy Content and Conversion Rates..... 40

Appendix III - Related Reports from USEU and MS Posts 40

I. Summary



Consumption

EU fuel bioethanol consumption surged 44 percent from 2020 to 2025 (see graph 1) due to an expanding gasoline pool and the rapid adoption of E10 fuel, transforming the European Union (EU) into a net importer. Growth was primarily driven by higher blending mandates, competitive fuel pricing, and increased consumer acceptance across major markets like Poland, France, and Germany. For 2026, fuel consumption is projected to rise by an additional 5.9 percent (6.4 percent in 2025), supported by favorable ethanol prices relative to gasoline and the ongoing rollout of the revised Renewable Energy Directive (REDII). However, this upward trajectory faces constraints from regional blending walls and

caps on crop-based bioethanol. In addition, high gasoline prices could shrink the gasoline pool and reduce the consumption of bioethanol.

Total EU Biomass-Based Diesel (BBD) consumption rose 4.1 percent in 2025 with a projected 2.8 percent increase in 2026. This growth is driven by rising greenhouse gas (GHG) emission reduction targets and new Sustainable Aviation Fuel (SAF) mandates. Overall BBD consumption faces pressures from a shrinking diesel fuel pool and increased transport electrification. Fatty Acid Methyl Ester (FAME) consumption is restricted by a seven percent volumetric blending limit and double-counting rules. While hydrogenation-derived renewable diesel (HDRD) and SAF hold a competitive advantage due to high GHG reduction values. A potential high Indirect Land Use Change (ILUC)-risk classification for soybean oil heavily threatens the EU market by cutting off major biodiesel imports from Argentina.

Production

Following a period of growth from 2020 to 2024, overall EU fuel bioethanol production is expected to remain stagnant throughout 2026 despite strong domestic demand and an abundant global grain supply. Leading producers like France and Germany face stabilized outputs near maximum capacities, while Hungarian production is constrained by a limited domestic corn supply and an import ban on Ukrainian grain. Expansion is primarily confined to Poland, and to a lesser extent, Czechia and Bulgaria, where favorable feedstock availability is supporting ongoing operations.

Total EU BBD production grew by over 3 percent in 2025 and is forecast to increase by another 4.2 percent in 2026. This upward trend is primarily driven by a 7.2 percent surge in HDRD and SAF production. Additionally, the anti-dumping duties on Chinese biodiesel created opportunities for increased domestic production. Although, despite expanding in 2025 and 2026, FAME production was negatively affected by a collapse in exports to the United States following changes to American tax credits. Feedstock utilization is shifting away from crop-based vegetable oils toward waste and residues, with Used Cooking Oil (UCO) experiencing significant growth. Conversely, the use of palm oil and soybean oil continues to decline rapidly due to impending sustainability requirements.

Trade

Based on the growing demand and stagnating domestic production, EU fuel bioethanol imports are forecast to further increase in 2026. Of the principal suppliers, probably only the United States and Ukraine will be able to expand their fuel bioethanol exports to the EU in 2026. The EU should continue to have access to ample U.S. bioethanol supplies throughout 2026 based on relatively flat U.S. domestic consumption and strong production. Ukraine has an EU TRQ of 158 million liters and has been ramping up bioethanol exports to the EU over the past decade. Brazil received improved access to the EU with the recent EU – Mercosur deal (see Policy & Programs chapter of this report). However, due to growing domestic demand, expansion of exportable supplies is forecast to be limited.

In 2025, EU BBD imports fell by 22 percent as new anti-dumping duties slashed Chinese supply. This import drop was initially softened by increases from Malaysia and South Korea. Early 2026 data show a total collapse in imports from Argentina alongside a rapid surge from India and Thailand. On the export side, diminishing trade with the United States occurred after the U.S. switched from a blenders' tax

credit to a producers' tax credit. Consequently, the United Kingdom became the EU's largest export market, experiencing a nearly 60 percent year-on-year increase that has continued to grow into early 2026.

Advanced Biofuels

Based on the minimum blending rates for “advanced” biofuels (as defined by the European Commission) produced with agricultural and forestry byproducts listed in Part A of Annex IX of the REDII, the consumption of these fuels must increase significantly by 2030. Most of the advanced biofuel demand is either HDRD produced from municipal and agricultural waste streams such as sewage sludge and fractions from palm oil mills, vegetable oil refineries, and pulp and paper plants, or bioethanol produced from food waste streams and fractions from celluloses production. With the introduction of catch and cover crops to Part A of Annex IX potentially a larger share of advanced biofuels will be produced with the use of these feedstocks.

A larger portion of biofuels is produced using waste oils and fats listed in Part B of Annex IX of the REDII. About a third of BBD (including HDRD) is produced from UCO and animal fats. The REDII sets a maximum consumption limit of 1.7 percent (energy-based) of all transport fuels can be biofuels produced with these waste oils and fats, but EU Member States can modify this limit, if justified, considering the limited availability of alternative feedstocks. Sourcing feedstocks from third countries could support a production expansion to keep these renewable transport fuels competitive.

II. Policy and Programs

The EU's Renewable Energy Directive (RED)

The [EU Energy and Climate Change Package](#) (CCP) ran from 2010-2020. The [RED](#), which was part of the CCP package, entered into force on June 25, 2009, and expired on December 31, 2020. The CCP required the EU to achieve a binding target whereby 20 percent of its overall energy use would be powered from renewable sources and 10 percent energy use in transportation for each Member State would come from renewables by 2020. For more information about RED, please see the [2020 Biofuels Annual Report](#).

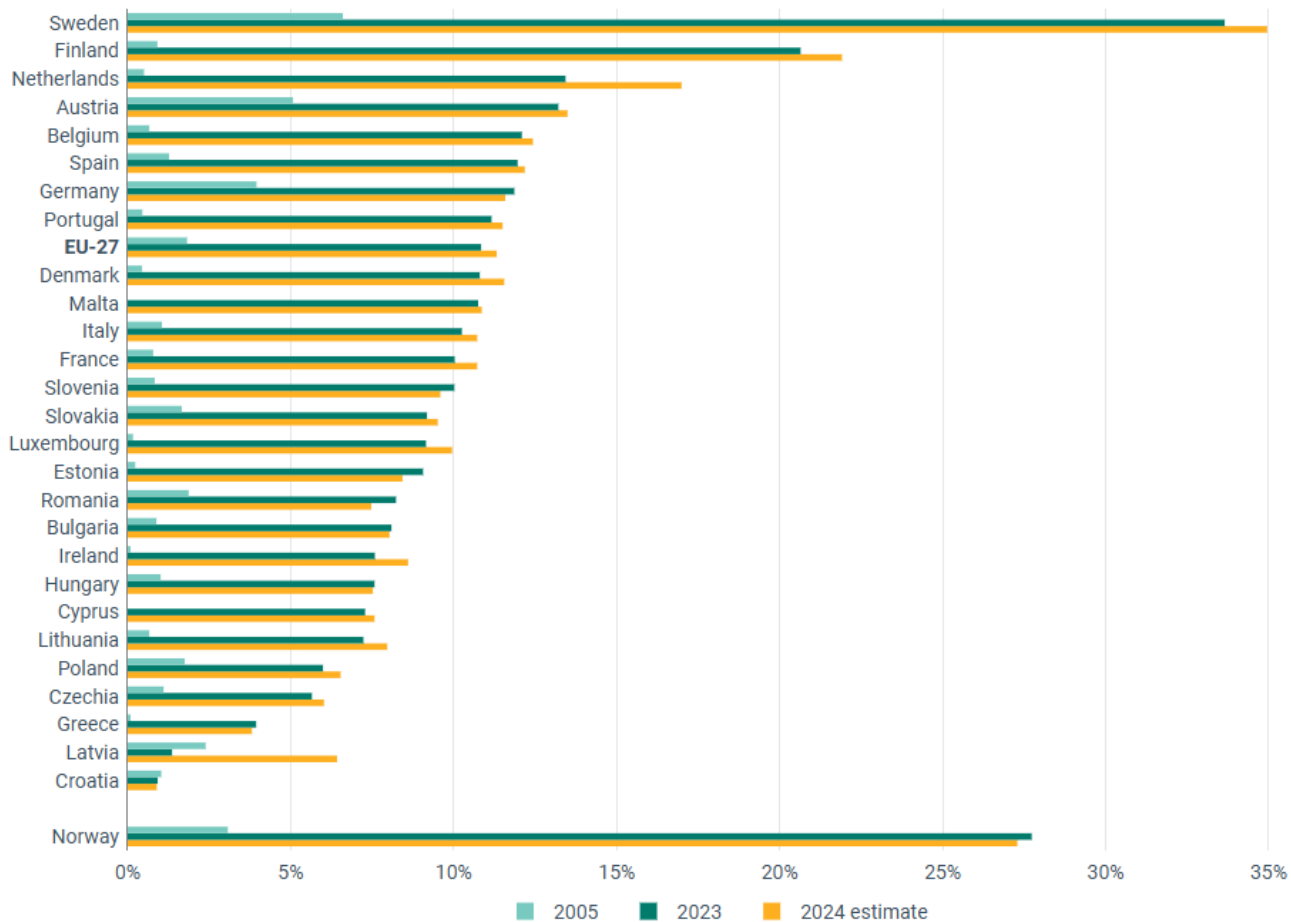
The Renewable Energy Directive II (REDII)

The European Union (EU) adopted the new REDII for the period 2021-2030 in 2018. Most of the provisions of [Directive 2018/2001](#) entered into force on January 1, 2021. The Directive was amended in October 2023 by [Directive 2023/2413](#) to align the REDII with the EU's Green Deal ambitions for a reduction in greenhouse gas emissions of 55 percent by 2030 (compared to 1990) and carbon neutrality by 2050. The revised REDII entered into application on May 21, 2025. Some Member States have not yet fully transposed the Directive into national law.

Uptake of Renewables in the EU

In 2024, the EU reached a 25.4 percent share of its final energy use from renewable sources, up 0.9 percent from 2023. The average share of energy from renewable sources in transport was 11.3 percent in 2024 (this includes double counting of advanced biofuels)¹.

Graph 2. Share of energy from renewable sources used in transport by EU Member State and Norway



The REDII Renewables Targets

The revised REDII sets out an overall renewable energy target of at least 42.5 percent binding at EU level by 2030. For transport, Member States can choose between a target of reducing greenhouse gas (GHG) intensity by 14.5 percent by 2030 (compared to 1990) or ensuring a share of at least 29 percent of renewables in final energy consumption by 2030. The Directive also sets out a binding target on non-crop based advanced biofuels of 1 percent in 2025 and 5.5 percent in 2030, of which a share of at least 1 percentage point is from renewable fuels of non-biological origin by 2030. The EU capped crop-based biofuels at the level consumed in each Member State in 2020, with an additional 1 percentage point

¹ Source: European Environment Agency, [Use of renewable energy for transport in Europe](#)

allowed over present consumption up to an overall cap of 7 percent. Member States can also set a lower limit for conventional biofuels than prescribed in the REDII.

For advanced biofuels, defined as biofuels made from feedstock listed in the following Table 1, the REDII introduces two different sets of targets for biofuels made from feedstocks listed in Part A of Annex IX and feedstock listed in Part B. Biofuels from feedstocks listed in Part A must be supplied at a minimum 1 percent in 2025 and 5.5 percent by 2030, of which a share of at least 1 percentage point is from renewable fuels of non-biological origin in 2030. Biofuels produced from feedstocks listed in Part B will be capped at 1.7 percent (energy-basis) in 2030 except in Cyprus and Malta. Please note that Member States may, where justified, modify that limit, considering the availability of feedstock. Any such modification shall be subject to approval by the European Commission (EC). Advanced biofuels can be double counted although not all Member States do this. In May 2024, the EC adopted [Delegated Directive \(EU\) 2024/1405](#) which adds new feedstock in Annex IX of the REDII. These feedstocks are in italics in the table below.

Table 1. Advanced Biofuel sources, Part A and Part B of Annex IX in the REDII

Part A	Part B
• Algae if cultivated on land in ponds or photobioreactors • Biomass fraction of mixed municipal waste • Biowaste from private households subject to separate collection • Biomass fraction of industrial waste not fit for use in the food or feed chain • Straw • Animal manure and sewage sludge • Palm oil mill effluent and empty palm fruit bunches • Crude glycerin • Bagasse • Grape marcs and wine lees • Nut shells • Husks • Cobs cleaned of kernels of corn • Biomass fraction of wastes and residues from forestry and forest-based industries • Other non-food cellulosic material • Other ligno-cellulosic material except saw logs and veneer logs • <i>Fusel oils from alcoholic distillation</i> • <i>Raw methanol from kraft pulping stemming from the production of wood pulp</i> • <i>Intermediate crops, such as catch crops and cover crops that are grown in areas where due</i>	• Used cooking oil • Some categories of animal fats • <i>Damaged crops that are not fit for use in the food or feed chain, excluding substances that have been intentionally modified or contaminated in order to meet this definition</i> • <i>Municipal wastewater and derivatives other than sewage sludge</i> • <i>Crops grown on severely degraded land excluding food and feed crops and feedstocks listed in Part A of this Annex, where not used for the production of biofuel for the aviation sector</i> • <i>Intermediate crops, such as catch crops and cover crops, and excluding feedstocks listed in Part A of this Annex, that are grown in areas where due to a short vegetation period the production of food and feed crops is limited to one harvest and provided their use does not trigger demand for additional land and provided the soil organic matter content is maintained, where not used for the production of biofuel for the aviation sector</i>

<i>to a short vegetation period the production of food and feed crops is limited to one harvest and provided their use does not trigger demand for additional land, and provided the soil organic matter content is maintained, where used for the production of biofuel for the aviation sector</i> • <i>Crops grown on severely degraded land, except food and feed crops, where used for the production of biofuel for the aviation sector</i> • <i>Cyanobacteria</i>	
--	--

A non-exhaustive list of waste and residues covered by Annex IX of REDII can be found in Annex IV of [Implementing Regulation 2022/996](#) on rules to verify sustainability and greenhouse gas emissions saving criteria and low indirect land-use change (ILUC)-risk criteria. The substances listed in that annex shall be considered as falling under a category of raw material set out in Annex IX without being explicitly mentioned.

In November 2025, the EU published Regulation 2025/2181 establishing new requirements for the imports of used cooking oil (UCO) containing or consisting of materials of animal origin, including a declaration by the importer of UCO intended for the EU. This Regulation will enter into force on [November 19, 2027](#).

Implementation of the REDII and Advanced Biofuels

Some EU MS plan to achieve higher blending rates of advanced biofuels than required by the REDII. For more information about the mandates see our FAS GAIN report: [Biofuel Mandates in the EU by Member State – 2025](#), published in January 2026. An update of this report is anticipated to be published in the summer of 2026.

The REDII Sustainability Criteria

To qualify towards the REDII targets, biofuels, bioliquids, and biomass consumed in the EU must comply with strict sustainability criteria provided in article 29 of the REDII. This article sets requirements on the minimum level of GHG savings, safeguarding against the conversion of high-carbon content lands and protection of biodiversity.

High-Risk Indirect Land Use Change (ILUC) Biofuels: The REDII introduces specific criteria for high-risk ILUC biofuels through [Delegated Act 2019/807](#), which determines high-risk ILUC biofuels. The EC defines high ILUC-risk feedstock as feedstock for which the share of expansion of the production into land with high-carbon stock is higher than ten percent since 2008 with an annual expansion of more than one percent. Given the calculations of the EC in 2019, only palm oil fell under this definition. The use of high-risk ILUC biofuels that count towards the REDII renewable energy targets, were capped at 2019 levels through 2023 and are scheduled to be phased out by 2030. The Delegated Act and [Implementing Regulation 2022/996](#) set out criteria for certifying low-risk ILUC biofuels, which were already defined in article 2 of the REDII.

In April 2026, the EC adopted a [delegated regulation updating the](#) methodology and data for high ILUC-risk biofuels. With the new proposed rule, the EC considered using baseline data from 2014 instead of 2008. Considering the new timeframe, and an update to the methodology in which high-risk ILUC feedstocks are measured, both palm oil and soybeans fell under the definition of a high ILUC-risk feedstock for biofuel use and therefore could not be able to count towards the EU's targets for biofuel use by 2030. On July 8, 2026, the European Parliament rejected the Commission's delegated act and called on the Commission to submit a revised delegated act incorporating a regionalization approach that accounts for differences in soy production regions. The timeline for the version of the delegated act is uncertain but a proposal is not expected before 2027.

The REDII Greenhouse Gas (GHG) Savings

The REDII introduces new compliance measures for GHG emission criteria for biofuels used in transport and counted towards targets. The EC is allowed to revise and update the default values of GHG emissions when technological developments make it necessary. Economic operators have the option to either use default GHG intensity values provided in the REDII or to calculate actual values for their pathway which must then be certified.

Table 3. GHG emission saving thresholds in the REDII for the transport sector

Plant operation start date	Transport biofuels	Transport renewable fuels of non-biological origin
Before October 2015	50%	70%
After October 2015	60%	70%
After January 2021	65%	70%
After January 2026	65%	70%

Compliance With Sustainability and GHG Emission Saving Criteria – Voluntary schemes

Voluntary schemes and national certification schemes of EU Member States help to ensure that biofuels, bioliquids, and biomass fuels are sustainably produced by verifying that they comply with the EU sustainability criteria. Following the entry into force of the REDII, voluntary schemes recognized under RED had to adjust the certification approaches to meet the new requirements. Those additional rules are enshrined in [Implementing Regulation 2022/996](#) which details the rules to verify sustainability and GHG emissions saving criteria and low ILUC change-risk criteria. This Regulation sets implementing rules to ensure economic operators comply with the sustainability criteria and provide accurate data on GHG emission savings of the REDII. More information about the recognition process can be found on the EC [website](#). The EC also made available an updated assessment [protocol](#).

Recognition by the EC is not a pre-requisite for certification. EU Member States may accept evidence from voluntary schemes or national certifications schemes set up by Member States not recognized by the EC if the competent authorities in those countries are confident about the quality of the certification services provided by these schemes.

Additional National Sustainability Requirements

The REDII allows Member States to establish additional sustainability criteria for biomass fuels. Before December 31, 2026, the EC is scheduled to assess the impact of such additional criteria on the internal market, accompanied, if necessary, by a proposal to ensure harmonization at EU-level. The REDII also allows Member States to set a limit lower than the seven percent allowed for biofuels, bioliquids, and

biomass fuels produced from food and feed crops. Member States can also distinguish between different biofuels, bioliquids, and biomass fuels produced from food and feed crops, considering the best available evidence on ILUC impact. EU Member States may, for example, set a lower limit for the share of biofuels, bioliquids, and biomass fuels produced from certain oil crops or ban them entirely as France has done with palm and soybean oil.

EU Database for Biofuels

The REDII required the EC to set up a Union Database for Biofuels (UDB) to enable the traceability of liquid and gaseous biofuels in the EU. This database was established in part to respond to product mislabeling (fraud) concerns over imports of certain biofuels, in particular biomass-based diesel made from UCO. It went live on January 15, 2024, but it is not yet mandatory for economic operators to digitally report their transactions when selling biofuels into the EU market. More information can be found on the [EC website](#).

The Fuel Quality Directive (FQD)

The [Fuel Quality Directive](#) required a reduction of the GHG intensity for transport fuels by at least six percent by 2020 (from 1990). It was amended by the revised REDII. This amendment removed the GHG intensity reduction target from the FQD and into the REDII. In addition, the FQD limits ethanol blends to ten percent or less when ethanol is used as an oxygenate in standard gasoline burning internal combustion engines (not applicable for flex-fuel engines).

Sustainable Aviation Fuels (SAF)

In October 2023, the EU adopted [Regulation 2023/2405](#) on sustainable aviation fuel (SAF). The Regulation requires aviation fuel suppliers to ensure all aviation fuel made available to aircraft operators at each EU airport contains an increasing minimum share of SAF, including a minimum share of synthetic aviation fuels (renewable fuels of non-biological origin), in accordance with the values and dates of application set below:

Table 4. Targets in the proposed SAF Regulation (volume based)

Date of application	Minimum share of SAF	Minimum share of synthetic fuels
January 1, 2025	2%	N/A
January 1, 2030	6%	1.2%
January 1, 2035	20%	5%
January 1, 2040	34%	10%
January 1, 2045	42%	15%
January 1, 2050	70%	35%

The EC defines SAF as aviation fuels that are either synthetic aviation fuels, recycled carbon aviation fuel, advanced biofuels composed of feedstocks listed in part A and B of Annex IX, or biofuels produced from feedstock other than food and feed crops which comply with the sustainability and GHG emissions criteria. The EU decided not to include first generation biofuels such as feed and food and crop-based biofuels for sustainability reasons. The EC defines synthetic aviation fuels as renewable fuels of non-biological origin.

Sustainable Maritime Fuels (SMF)

In September 2023, the EU adopted [Regulation \(EU\) 2023/1805](#) on the use of renewable and low-carbon fuels in maritime transport. This Regulation aims to promote the use of renewable and low-carbon fuels and alternative energy sources in maritime transport throughout the EU. To achieve this, it sets a limit on the intensity of GHG emissions from the energy used on board, which becomes stricter over time. First generation biofuels such as feed and food and crop-based biofuels cannot be used to reach the targets set by the Regulation. However, advanced biofuels and renewable fuels of non-biological origin comply with sustainability criteria and GHG emissions laid down in REDII can count towards the targets.

EU Climate Law

On July 9, 2021, Regulation 2021/1119, also known as the [EU Climate Law](#), was published in the EU Official Journal. The Climate Law enshrines a legally binding target of net zero GHG emissions by 2050. EU Institutions and Member States are bound to take the necessary measures at EU and national level to meet the target. The Climate Law includes measures to keep track of progress and adjust the EU's actions accordingly. The text also includes a reduction of net GHG emissions by at least 55 percent by 2030, compared to 1990 levels.

In March 2026, the EU adopted an amendment to the European Climate Law to set an [EU climate target for 2040](#). With [Regulation 2026/667](#), the EU adopted a legally binding headline emission reduction target of 90 percent by 2040, relative to 1990, with a domestic target of 85 percent and up to 5 percent of international carbon credits.

New CO₂ emissions standards for vehicles

In April 2023, the EU adopted [Regulation 2023/851](#) which required a 100 percent reduction target for CO₂ tailpipe emissions from new passenger cars and new light commercial vehicles by 2035. This Regulation effectively bans the sale of new internal combustion engine passenger cars and vans by 2035. The agreed text sets intermediate 2030 targets at 55 percent fleet-wide CO₂ emissions reduction (compared to 2021 levels) for new cars and a 50 percent reduction for vans.

In December 2025, the EC [proposed](#) a review of the regulation setting CO₂ emission standards for passenger cars and vans. The EC is proposing to postpone beyond 2035 the carbon neutrality target for EU cars and vans. The revised rules would lower the reduction target for CO₂ tailpipe emissions to 90 percent in 2035 (compared to 2021) for passenger cars and for vans to 40 percent in 2030 and 90 percent in 2035. In June 2024, the EU adopted [Regulation 2024/1610](#) revising CO₂ emission standards for heavy-duty vehicles. Under the Regulation, CO₂ emissions must be reduced on average compared to 2019 levels by 45 percent by 2030, 65 percent by 2035 and 90 percent by 2040 onwards. The new Regulation also sets a 100 percent zero-emission target for urban buses by 2035 and an intermediate target of 90 percent by 2030.

Revision of the Energy Tax Directive

The EC also announced a revision of the [Energy Tax Directive](#) (ETD). The EC noted that biodiesel, and especially ethanol, are disadvantaged by the volume-based taxation (rates expressed per liter), because one liter of these fuels has a lower energy content than one liter of the fossil fuels they replace while the

same tax rate applies. Therefore, the EC proposed to set different minimum levels of taxation applicable to fossil fuels and biofuels on an energy basis (€/gigajoule). This would be accompanied by a transition period for food and feed crop biofuels and low-carbon fuels. The proposal continues through the legislative process and has not yet been adopted.

Table 5. Proposed minimum level of taxation applicable to motor fuels (in €/gigajoule)

Motor Fuel	Start of transitional period (01/01/2023)	Final rate after completion of transitional period (01/01/2033)
Petrol	10.75	10.75
Gasoil	10.75	10.75
Sustainable food and feed crop biofuels	5.38	10.75
Sustainable biofuels	5.38	5.38
Low-carbon fuels	0.15	5.38
Advanced sustainable biofuels and biogas	0.15	0.15
Renewable fuels of non-biological origin	0.15	0.15

EU Deforestation-free Supply Chain Regulation (EUDR)

In June 2023, the EC adopted [Regulation 2023/1115](#), also known as the EU Deforestation-free Supply Chain Regulation (EUDR), aimed to prevent products causing deforestation from entering the EU. The proposal targets products identified as main deforestation drivers, including palm oil and soybeans. In December 2025, the EU published [Regulation 2025/2650](#) which delayed the EUDR's entry into application for another year until December 30, 2026. The Regulation also creates new definitions that provide simplifications to some, primarily EU, operators. For more information about the changes introduced by Regulation 2025/2650, please see GAIN Report: [EC Publishes Simplification Review Report on Deforestation Regulation](#).

The EU Taxonomy for Sustainable Activities

In order to meet the EU's climate targets for 2030 and reach the objectives of the European Green Deal, the EC adopted the [Taxonomy Regulation](#) in June 2020. This Regulation establishes the framework for an EU taxonomy for sustainable activities by setting out four overarching conditions that an economic activity must meet to qualify as 'environmentally sustainable'. The Taxonomy Regulation aims to act as a screening mechanism to define sustainable activities to steer private investment to activities the EC deems sustainable. It creates three different categories: "sustainable activities", "transitional activities" and "enabling activities." The EC classifies crop-based biofuels as sustainable activities. More information can be found in GAIN Report: [Commission Adopts Taxonomy for Green Investments](#).

Policy Response to the War in Ukraine

The EU granted Ukraine a temporary liberalization of trade with its Autonomous Trade Measures (ATMs) Regulation, which was in effect from June 4, 2022, until June 5, 2025. This regulation

suspended import duties, quotas, and trade defense measures on Ukrainian exports to the EU, providing significant economic support during the Russia-Ukraine war. Between June 6 and October 29, 2025, the EU-Ukraine trade relationship reverted to the 2014 Deep and Comprehensive Free Trade Area (DCFTA) via [Regulation 1132/2025](#) reintroducing import quotas on Ukrainian agricultural goods. On October 29, 2025, the revised EU-Ukraine DCFTA entered into force ([Regulation 2025/2199](#)).

The revised DCFTA increases market access in both directions compared to the 2014 agreement, but limits EU imports of sensitive agricultural products compared to the levels under the ATMs. It also enshrines a new safeguard clause and provides for the alignment of Ukrainian and EU production standards. The new DCFTA includes a TRQ for Ukrainian ethanol of 100,000 MT (expressed in net weight). For more information, please see GAIN Report [European Union Revises Import Quotas for Agricultural Products from Ukraine](#).

Market Access

The EU is negotiating and has implemented several Free Trade Agreements (FTAs) with other countries and regions, which include concessions on biofuels. Additional information is available on the [EC website](#).

Mercosur: On January 17, 2026, the EU and the four Mercosur countries - Argentina, Brazil, Paraguay, and Uruguay - formally signed the EU-Mercosur Partnership Agreement (EMPA) and a parallel Interim Trade Agreement (iTA). To address long-standing concerns regarding market disruption, the European Parliament approved a reinforced bilateral safeguard regulation on February 10, 2026. This mechanism allows the EU to temporarily suspend tariff preferences or reinstate duties if a surge in agricultural imports causes or threatens serious injury to EU producers. A five percent increase in imports of sensitive products over a three-year average can now trigger a formal investigation. The iTA was applied provisionally on 1 May 2026, and tariffs were removed from certain products as of day one. Under the agreement, duties will be phased out on 91 percent of EU exports to Mercosur and 92 percent of Mercosur exports to the EU. Mercosur countries will be allocated new TRQs with a five-year phase in period (six annual stages). For bioethanol, this includes:

- 450,000 MT of ethanol used by the chemical industry.
- 200,000 MT of ethanol for all other uses, including biofuels.

Mexico: On January 17, 2025, the EU and Mexico announced the conclusion of a new FTA. In September 2025, the EC adopted proposals for an Interim Trade Agreement (iTA) to fast-track the trade pillar while the broader agreement undergoes full member-state ratification. As of February 2026, the agreement is moving toward a formal signature expected before the summer, with implementation benefits targeted for later this year or early 2027. The agreement abolishes customs duties for most goods. Additionally, Mexico will be allocated new TRQs with a 5-year phase in period, including a TRQ for 5,500 MT for ethanol for energy use.

Duties

[Regulation 2017/2321](#) defines the EU's anti-dumping and anti-subsidy rule. Duty rates for fuels are listed below; for a historical discussion of how EU harmonized system (HS) customs codes have changed and influenced trade please see the [EU Biofuels Annual 2017](#).

Table 6. Most-Favored Nation (MFN) Duty Rates for Biofuels

HS Code	Description	Duty Rate
38260010	FAME above 96.5% and up to 100% by volume	6.5%
38260090	FAME greater than 30% and up to 96.5% by volume	6.5%
271020	Petroleum oils containing FAME up to 30% by volume	3.7%
220710	Undenatured ethanol	€19.2/hl
220720	Denatured ethanol	€10.2/hl
27101942	Petroleum oil having a bio-based carbon content of at least 80 % by weight	3.5% ²

Antidumping (AD) and Countervailing (CV) Duties Against U.S. Biodiesel

For more information, please see the [Biofuels Annual of 2025](#) and GAIN Report: [EU Extends Its Anti-Dumping Duty and Countervailing Duties on Imports of US Biodiesel](#). On September 12, 2025, the EU published [Implementing Regulation 2025/1869](#) to clarify that sustainable aviation fuels (SAF) are in the scope of Regulation 2021/1266.

Biodiesel AD and CV Duty Actions Against Argentina and Indonesia

On May 5, 2025, the EU published [Implementing Regulation \(EU\) 2025/835](#) imposing a definitive countervailing duty on imports of biodiesel from Argentina for another 5 years. The rates of the definitive countervailing duty range between 25 and 33.4 percent depending on the company.

In December 2019, the EU imposed countervailing duty on imports of biodiesel from Indonesia with [Implementing Regulation 2019/2092](#). The CV duty ranges from 8 to 18 percent depending on the company. On March 3, 2026, the EU renewed the countervailing duties on Indonesian biodiesel for another 5 years with [Implementing Regulation 2026/479](#). For more information, please see the [Biofuels Annual of 2025](#).

Biodiesel Antidumping Actions Against China

On February 10, 2025, the EC imposed antidumping duties on Chinese biodiesel imports ([Implementing Regulation 2025/261](#)). The measures had been [applied provisionally](#) since August 14, 2024. The measures are imposed on biodiesel in pure form or as included in a blend but exclude aviation biofuels known as SAF. The EU imposed AD duties on Chinese biodiesel range between 10.0 and 35.6 percent depending on the company.

Suspension of the GSP+ Tariff Preferences for Ethanol Originating in Pakistan

On June 20, 2025, the EC suspended Pakistan's Generalized Scheme of Preferences Plus (GSP+) status for ethanol imports, effective June 20, 2025, following the publication of [Implementing Regulation 2025/1206](#). The GSP+ status provides preferential access to the EU market for certain products by removing or reducing import duties. This suspension, which applies for two years, was initiated due to concerns about the impact of Pakistani ethanol exports on the EU market. Fuel-grade ethanol is not included in the measures.

² Please note that the duty is suspended, on an autonomous basis, for an indefinite period for gas oil having a Sulphur content not exceeding 0.2 % by weight.

Surveillance of Imports of Renewable Ethanol for Fuel

On September 15, 2023, the EC [introduced](#) retroactive surveillance measures on imports of bioethanol for fuel from several countries, including the United States. The measures were launched after imports of bioethanol into the EU increased by close to 80 percent between 2021 and 2022. The most important exporting countries in terms of volumes in 2022 were Brazil, the United States, the United Kingdom, and Peru. The introduction of the surveillance measures is the first step that can be taken before the introduction of possible antidumping or countervailing measures.

III. Ethanol

Bioethanol is produced by fermenting the carbohydrate components of plant materials. In the EU, the most used feedstocks are grains (e.g., corn, other coarse grains, and wheat kernels) and sugar beet. ‘Synthetic’ ethanol made from petroleum fuels is restricted to a very small market and is not included in this report.

EU Production, Supply and Demand Table

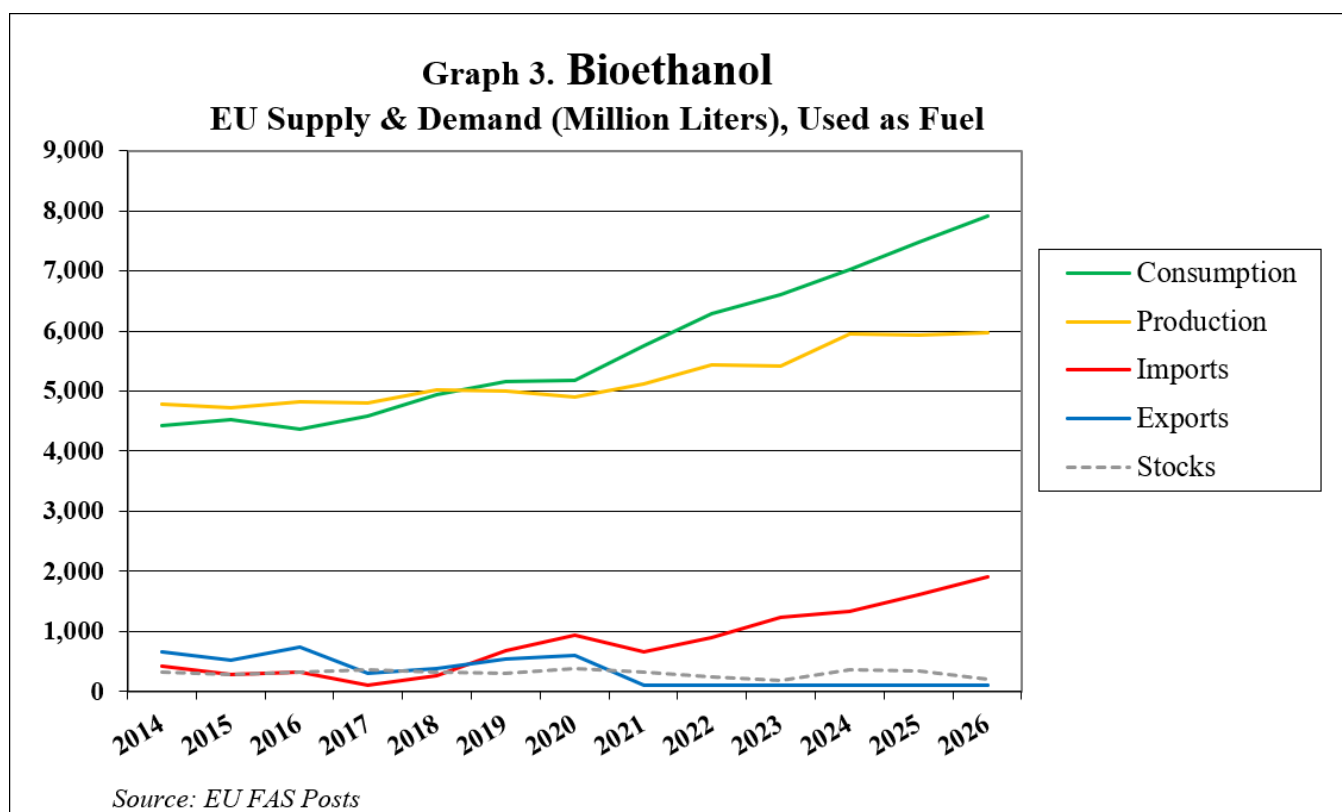
Table 7. Ethanol Used as Fuel and Other Industrial Chemicals										
(Million Liters)										
Calendar Year	2017	2018	2019	2020	2021 ^r	2022 ^r	2023 ^r	2024 ^r	2025 ^e	2026 ^f
Beginning Stocks	359	384	358	330	430	344	276	211	389	362
Fuel Begin Stocks	325	355	332	304	385	318	253	193	368	342
Production	5,399	5,565	5,549	5,841	5,690	5,944	5,817	6,431	6,382	6,421
Fuel Production	4,806	5,015	5,000	4,909	5,119	5,437	5,408	5,956	5,938	5,975
<i>of which cellulosic (a)</i>	10	10	10	35	70	70	90	20	20	20
Imports	881	800	1,100	1,491	1,125	1,985	1,960	1,870	1,914	2,100
Fuel Imports	110	270	681	941	670	894	1,242	1,341	1,604	1,911
<i>of which ETBE (b)</i>	9	9	14	26	19	82	115	145	133	130
Exports	349	426	592	647	150	150	150	150	150	150
Fuel Exports	299	376	542	597	100	100	100	100	100	100
Consumption	5,906	5,965	6,085	6,585	6,751	7,847	7,692	7,973	8,173	8,496
Fuel Consumption	4,587	4,932	5,167	5,172	5,756	6,296	6,610	7,022	7,468	7,911
Ending Stocks	384	358	330	430	344	276	211	389	362	237
Fuel Ending Stocks	355	332	304	385	318	253	193	368	342	217
Refineries Producing First Generation Fuel Ethanol (Million Liters)										
Number of Refineries	57	56	52	54	58	53	55	53	53	53
Nameplate Capacity	7,418	7,284	7,271	7,478	8,070	8,108	8,332	8,380	8,563	8,582

Capacity Use (%)	65	69	69	65	63	66	64	71	69	69
Refineries Producing Cellulosic Fuel Ethanol (Million Liters)										
Number of Refineries	1	1	1	2	3	3	2	1	1	1
Nameplate Capacity	10	10	10	40	110	110	100	30	30	30
Capacity Use (%)	100	100	100	88	64	64	90	67	67	67
Co-product Production (1,000 MT)										
DDGs	3,307	3,518	3,441	3,608	3,658	4,048	4,116	4,499	4,426	4,538
Corn Oil	144	186	198	188	192	215	211	227	222	225
Feedstock Use for Fuel Ethanol (1,000 MT)										
Wheat Kernels	3,955	3,125	2,721	3,013	2,581	2,798	3,048	3,679	3,802	3,862
Corn Kernels	4,967	6,410	6,819	6,497	6,613	7,403	7,284	7,828	7,657	7,753
Barley Kernels	391	485	443	518	652	625	692	1,057	1,052	1,145
Rye Kernels	518	487	233	445	589	433	169	143	146	146
Triticale	736	733	778	1,052	1,252	1,673	1,957	1,668	1,485	1,591
Sugar Beets	7,768	7,025	7,352	4,806	6,843	4,724	3,401	4,410	5,115	4,202
Cellulosic Biomass	40	40	40	140	280	280	360	80	80	80
Market Penetration (Million Liters)										
Fuel Ethanol Use	4,587	4,932	5,167	5,172	5,756	6,296	6,610	7,022	7,468	7,911
Gasoline Pool 1/	91,254	96,289	98,433	86,222	93,938	99,326	102,888	107,333	110,141	112,699
Blend Rate (%)	5.03	5.12	5.25	6.00	6.13	6.34	6.42	6.54	6.78	7.02

Sources/Notes: r = revised / e = estimate / f = forecast of EU FAS Posts. Footnote :1/ Fuel pool defined as gasoline plus all biocomponents (ethanol, ETBE, methanol). Source: IEA, Oil Market Report, June 2026.

Production capacity as of December 31 of year stated. Ethanol use: Eurostat statistics and FAS Posts projections.

Trade and stocks data: See Notes section. Footnotes: (a) Production is an estimate. For more information see section Advanced Biofuels. (b) ETBE HS code 29091910, ETBE contains 45 percent ethanol which is the volume reported. (c) From 2021 to 2026, EU fuel bioethanol and total bioethanol exports are anticipated to be minimal at estimated at respectively 100 million liters and 150 million per year. (d) Calculated co-product production (theoretical maximum) based on estimated feedstock use in fuel bioethanol production.



Consumption

Table 8. EU Fuel Bioethanol Consumption By Member State (Million Liters)								
Calendar Year	2019^r	2020^r	2021^r	2022^r	2023^r	2024^e	2025^e	2026^f
Germany	1,435	1,378	1,467	1,508	1,581	1,622	1,645	1,685
France	1,239	1,087	1,378	1,648	1,625	1,415	1,448	1,455
Poland	372	359	409	456	478	659	722	760
Netherlands	366	430	444	477	505	504	506	525
Belgium	228	223	277	296	324	335	342	342
Sweden	178	187	229	297	371	267	316	335
Spain	257	195	248	224	306	304	285	315
Hungary	189	167	180	184	161	171	180	185
Other	903	1,146	1,124	1,206	1,259	1,745	2,024	2,309
Total	5,167	5,172	5,756	6,296	6,610	7,022	7,468	7,911

r = revised / e = estimate / f = forecast EU FAS Posts. Source : EU FAS Posts and Eurostat

As from 2022, non-fuel use of ethanol, for industrial use and human consumption, is anticipated to have declined in the EU based on the lower use for sanitary purposes since the end of the large-scale COVID-19 outbreak and the contraction of the chemical sector^[60]. However, starting in 2019, consumption began to outpace production, resulting in the EU becoming a net importer of fuel bioethanol (see Graph 3

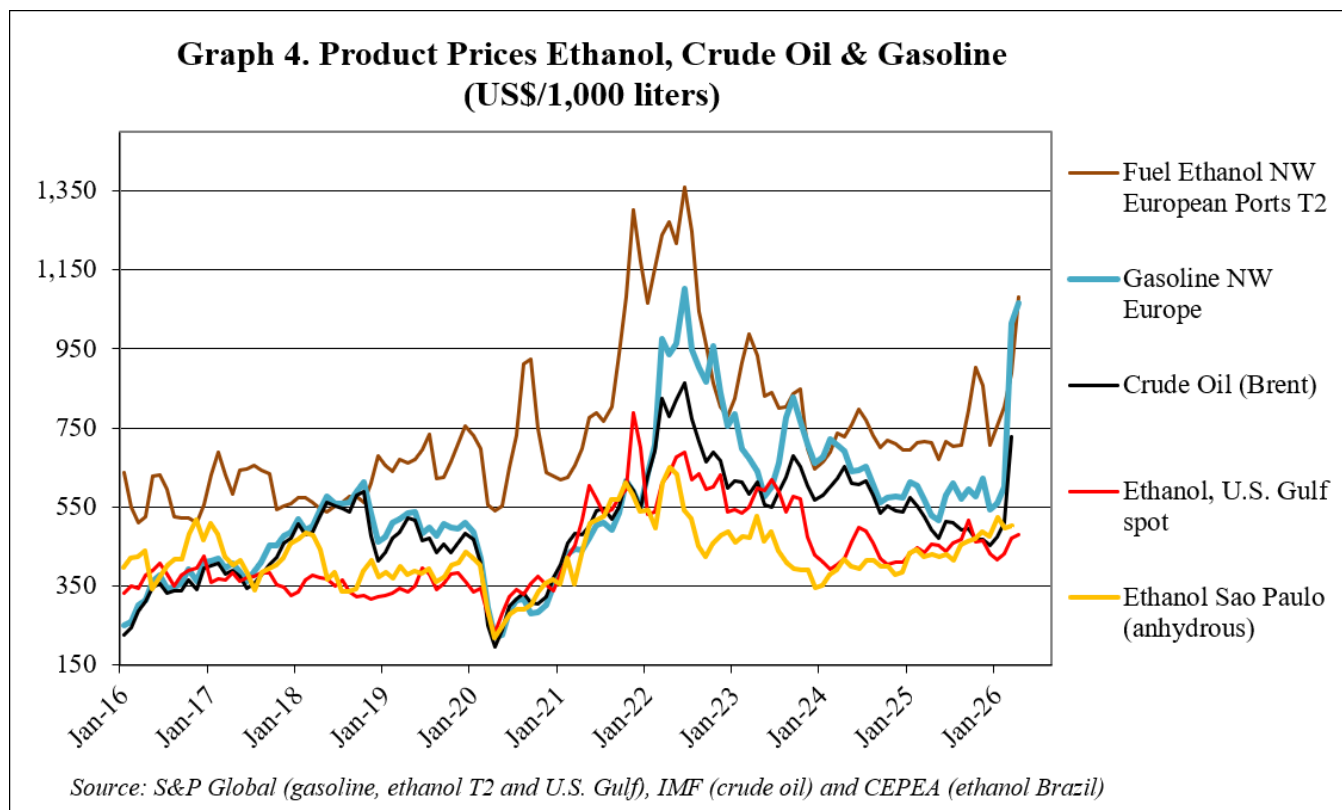
above). EU fuel bioethanol consumption has been expanding since 2016, though only with a slight increase in 2020 due to the transport lockdowns during the COVID-19 pandemic. After the lockdowns were lifted EU fuel ethanol consumption took off and increased by 44 percent during 2020 - 2025. Based on continued market expansion of E10 in many countries, FAS EU posts estimate an overall EU growth in bioethanol fuel use of 6.4 percent in 2025. Currently, E10 is available in nineteen EU Member States: Austria, Belgium, Bulgaria, Czechia, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Latvia, Lithuania, Luxembourg, the Netherlands, Poland, Romania, Slovakia, and Sweden (source ePURE). EU Member States which have not introduced E10 are mainly located in the Mediterranean region (Spain, Portugal, Italy, Slovenia, Croatia, Greece, Cyprus, and Malta). In Spain, E10 is permitted but E5 is the standard blend in petrol stations. In many EU Member States both E5 and E10 are available, and a gradual shift to the latter is taking place due to the increasing number of stations providing E10, further supported by the lower price of E10, growing acceptance by drivers, and a better adaptability of the car engines to higher ethanol blends. Another factor for the increased fuel bioethanol consumption through 2020 – 2025 is the overall increase in the gasoline fuel pool across the EU (source: International Energy Agency). Two factors play a key role in the rising gasoline consumption; economic growth boosting road transportation and the substitution of diesel by gasoline cars and gasoline hybrid cars. The increase of fuel bioethanol consumption during 2025 was mostly driven by Poland, France, Germany, and Sweden.

In Poland, the main driver for the expansion of bioethanol use is the increase of the blending mandates, from 3.20 percent in 2024 to 4.59 percent in 2025. In addition, E10 is gaining popularity after it was introduced in the beginning of 2024. The expanding gasoline pool is another important factor in the growing consumption of bioethanol in Poland. Similarly in France and Germany, bioethanol consumption increased based on an expansion of the gasoline pool and elevated sales of E10. In France, E10 reached a record share of 63³ percent on total gasoline use in December 2025 and consumption of E85 reached an estimated market share of six percent in April 2026. Germany's introduction of E10 in 2010 was met with skepticism due to reports it could harm the engines. But the increasing price discount between E5 and E10 is shifting consumer choice to the higher blend. In Germany, the market share of E10 increased from 27.5 percent in 2024 to 28.6 percent in 2025. In Sweden, the increase of the GHG reduction mandate from six to ten percent in 2025 drove a higher blending of bioethanol. Like France, Sweden also has a market for E85, which is incentivized through tax exemptions outside the mandate.

In 2026, EU bioethanol consumption is affected by the implementation of the revised second EU Renewable Energy Directive (REDII). However, implementation within the majority of the 27 member states is delayed, including France, Belgium, Poland, and Spain. The effect of the implementation is difficult to assess as each EU Member State will enforce the directive in a unique way and even the implications at the country level are often uncertain. Some EU Member States changed their mandates from caloric-based blending to GHG emission reduction. In some countries, another complicating factor is the abolishment of double counting by which, depending on the GHG savings of the blended bioethanol, a higher volume is needed to fulfill the mandates. Based on Post's assessments, the enforcement of the revised REDII will support a further expansion of bioethanol use in the EU in 2026. The blending wall of ten percent (of E10) and the crop cap (seven percent or lower) are, however, setting a maximum boundary for growth. For example, Spain and Portugal have a crop cap of 2.6 and 3.1 percent, respectively. For more information about the blending mandates see our FAS GAIN report:

³ Link in French [Données Superéthanol-E85](#)

[Biofuel Mandates in the EU by Member State – 2025](#), published July 16, 2025. An update of this report is anticipated to be published in the summer of 2026.



Besides the revision of the REDII, EU consumption of bioethanol in 2026 will also depend on the price in relation to that of gasoline. Since March 2026, ethanol prices have been relatively low compared to gasoline prices, supporting high bioethanol blends (see graph 4). In the EU, the bioethanol price (T2) is typically not related to feedstock prices but generally more linked to the fossil fuel prices. Nevertheless, the logistical situation in the Strait of Hormuz affects bioethanol prices less than gasoline prices, giving the former a price advantage. For example, in France, a renewed interest in E85 conversion kits has been reported as the E85 price level has been about half the price of conventional gasoline. E85 represents around a third⁴ of all bioethanol consumed in France. High gasoline prices could however curb total EU fuel demand. This effect has already been reported in France with increased teleworking, car sharing, and rising sales of train tickets and electric cars. The French government introduced a €70 million sectoral aid package to cushion the impact of rising fuel costs on transport, agriculture, fishing, and construction. In some EU Member States a reduction of bioethanol consumption is forecast, such as in Romania and Ireland. Recently the Romanian government amended the specific legislation and allowed fuel companies to reduce the blending mandate through April to June 2026 from eight to two percent as the minimum blending requirement. In its justification, the government cited difficulties in the supply chain related to the logistical complications in the Middle East.

⁴ Link in French [02-2026_CP-CONFERENCE-COLLECTIVE-BIOETHANOL-VF.pdf](#)

Overall, FAS Post projects the EU fuel bioethanol consumption to increase by 5.9 percent in 2026. The current blending of bioethanol is estimated at 7.0 percent. The market for bioethanol in aviation and shipping has potential for the medium to long term but is not anticipated to take off in 2026.

Production and Capacity

Table 9. EU Fuel Bioethanol Production
By Member State (million liters)

Calendar Year	2019^r	2020^r	2021^r	2022^r	2023^r	2024^e	2025^e	2026^f
France	1,103	975	962	1,111	1,124	1,119	1,076	1,090
Germany	676	700	738	759	727	797	805	805
Belgium	620	620	633	649	649	658	660	660
Netherlands	519	481	519	519	570	608	658	658
Hungary	642	629	662	632	590	765	633	633
Poland	286	276	338	404	429	543	604	633
Spain	547	501	553	497	547	554	546	550
Austria	254	222	246	259	246	238	240	240
Other	353	505	468	607	526	674	716	706
Total	5,000	4,909	5,119	5,437	5,408	5,956	5,938	5,975

r = revised / e = estimate / f = forecast EU FAS Posts. Source : EU FAS Posts and Eurostat

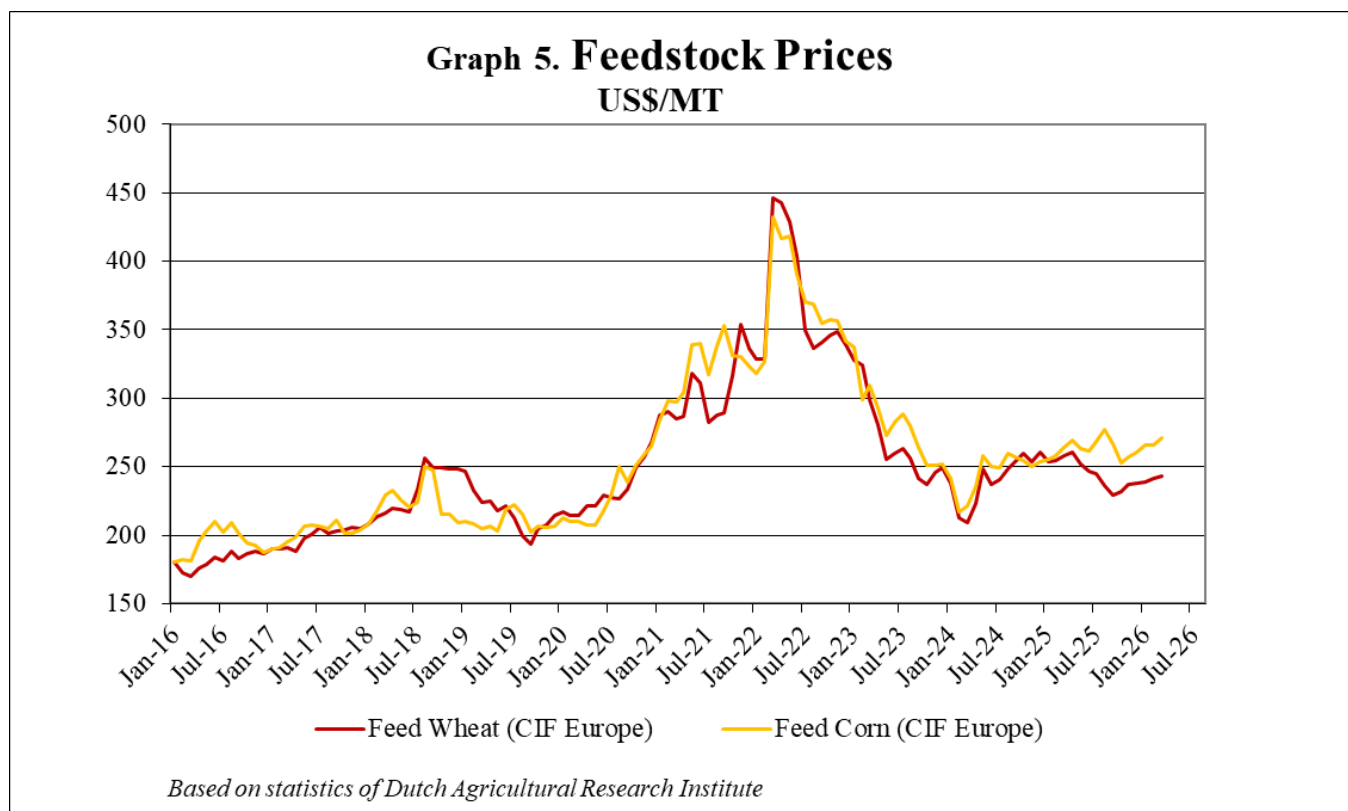
Overall EU fuel bioethanol production increased between 2020 – 2024 mainly driven by growing demand and the restriction of imports from third countries. But EU production is anticipated to have stagnated in 2025 and is forecast relatively flat this year, despite the strong domestic demand and ample supply of grains. In 2024, fuel bioethanol production increased most significantly in Hungary and Poland. But in 2025 and 2026, Hungarian producers experienced limited corn supplies, the main feedstock for bioethanol production in Hungary (see the next paragraph about feedstock use). In addition, Hungary has maintained a unilateral import ban on grain imports from Ukraine since September 16, 2023.

Since 2024, significant production expansion is only anticipated in the Netherlands and Poland, and to a lesser extent, Czechia and Bulgaria. Polish fuel bioethanol production increased continuously since 2020 driven by a further increase of gasoline demand and high availability of corn in Central Europe. Polish production is forecast to further grow this year. Czech fuel bioethanol production is estimated at 130 million liters in 2025, up 25 percent from 2024, reflecting stronger use of both grain and sugar-based feedstocks. Production is forecast to ease to 120 million liters in 2026 which is still well above the 2020–2024 average. In Bulgaria, production is forecast to grow by about 3-4 percent annually during 2025 and 2026. The main feedstock is corn, followed by wheat and sorghum.

Production by the two main producers in the EU, France and Germany, is projected to stabilize as no new plants will be opened and production is near maximum levels. Although production of fuel grade bioethanol could expand at the expense of bioethanol for industrial uses. Production in Western Europe, mainly the Netherlands, Belgium, and Spain may also depend on the global availability of corn and potential restrictions on imports from the United States. As of the drafting of this report, FAS Posts anticipate that these EU Member States will be able to source enough corn to maintain their production

levels (for more information see the next section).⁵ The logistical situation in the Middle East is not anticipated to affect EU bioethanol production. Stable feedstock prices and rising ethanol prices support producer margins. But, on the other hand, elevated gas prices reportedly could significantly undercut profits.

Feedstock Use and Production of Co-products



In the EU, nearly all bioethanol is produced from grains and sugar beet derivatives. Wheat is predominantly used in Germany and France. In Central Europe, particularly in France, Hungary, and Poland, the domestic corn supply is the basis of bioethanol production. Corn is also the preferred grain in the Netherlands, Belgium, and Spain, where most bioethanol plants are located at seaports, and corn is sourced from third countries (predominantly corn from Ukraine and the United States).

Even though corn and wheat prices are commonly at about the same level (see graph above), margins for producing corn ethanol are reportedly better than for producing wheat ethanol, which could further support the use of corn in the EU. The global supply of corn is anticipated to be abundant and favors its use for bioethanol production in Western Europe. There is an incentive to use non-genetically engineered (non-GE) corn as bioethanol producers in northwestern Europe prefer to market their distillers dried grains (DDG) as non-GE for the domestic feed market. The use of barley as feedstock for bioethanol production is forecast to further increase in 2026, mainly driven by the expanding use in Hungary. Barley is reportedly better adapted to severe droughts than corn.

⁵ For additional information regarding the EU's Grain Market, consult the latest EU's [Grain and Feed Report](#).

In France, Germany, Czechia, Belgium, and Austria, sugar beets and their derivatives are also used to produce bioethanol. In France, sugar beets are only processed for bioethanol in sugar beet processing plants that have on-site ethanol distillation capacity. In some other EU Member States, like Austria and Belgium, beet pulp or concentrated juice may serve as a feedstock for bioethanol production. The use of sugar beets for bioethanol increased in 2025, supported by the good EU harvest, and rose particularly in Germany and Czechia. In 2026, the production of sugar beet bioethanol is forecast to fall back to average levels. In Czechia this reflects the longer-term structural pressure on the Czech sugar sector and the reduced number of sugar beet processing facilities. For more information see the FAS GAIN [EU Grain and Feed Annual](#), published on April 20, 2026, and the FAS GAIN [EU Sugar Annual](#), published April 20, 2026.

In the EU, to reach the estimated 2026 production of 5.98 billion liters of fuel bioethanol, the required cereals volume needed is estimated at 14.5 MMT, an increase of about 360,000 MT compared to 2025. This is roughly 5.2 percent of total EU cereal production. Co-products from the bioethanol production process are distillers dried grains (DDG), wheat gluten, corn oil, and yeast concentrates. In 2026, the maximum theoretical production level (calculated, using the conversion factors listed at the end of this report) of DDG is forecast to reach 4.54 MMT, an increase of roughly 112,000 MT from 2025. This accounts for 2.9 percent of total EU feed grain consumption. The volume of sugar beets used to produce fuel bioethanol is estimated at 5.12 MMT in 2025, and 4.20 MMT in 2026. This is roughly 4.4 percent of total EU sugar beet production in 2026.

Trade

U.S. bioethanol exports to the EU are subject to the Most Favored Nation (MFN) import tariff and sustainability requirements when fuel ethanol is shipped. For more information see the Policy and Programs Chapter of this report and the [EU Biofuels Annual of 2020](#). Since 2022, EU fuel bioethanol imports surged and have remained at relatively high levels up to the publication of this report. Based on the methodology described in Chapter VIII – Notes on Statistical Data, 2025 EU fuel bioethanol imports are estimated at 1.60 billion liters. Since 2021, EU imports of bioethanol as ethyl-tert-butylether (ETBE) increased significantly from 19 million liters to 133 million liters in 2025. Note that any “light (petroleum) oils” found under HS code 2710.12 (gasoline) that may contain ethanol are not included in the balance or trade estimates. This is because preblended gasoline volumes arriving or leaving the EU are deemed to be small, and thus not appreciably affecting the balance.

Since 2024, the United States is the principal supplier of fuel bioethanol to the EU followed by Canada, Brazil, Peru, Guatemala, and Ukraine. In 2024 and 2025, the United Kingdom was the second and third largest supplier of ethanol (for all uses) to the EU, but the share of transport fuel is uncertain. Likewise, imports from Pakistan are not counted as bioethanol for fuel. Of these six top-suppliers likely only the United States and Ukraine will be able to expand their fuel bioethanol exports to the EU this year. The EU should continue have access to ample U.S. ethanol supplies throughout 2026 based on relatively flat U.S. domestic consumption and strong production. As of this report’s publication, it is uncertain if E15 will be introduced in the United States. But even if the proposed E15 bill were made a law, only a limited adoption is anticipated in 2026 because the bill allows the sale of E15 but does not mandate E15. Additionally, there will need to be infrastructure upgrades at many pumps to be able to sell E15, so the

rollout will likely not be immediate unless volumes are mandated or subsidies are offered to upgrade fueling stations. The United States commonly exports denatured ethanol to the EU, with the Netherlands as the main receiver, as this is covered by a lower import tariff. However, under implementation of the revised REDII, the Dutch Government has stated that only undenatured bioethanol will be eligible to meet REDII requirements as of January 2026. Consequently, U.S. suppliers will have to switch from exporting denatured to undenatured bioethanol to the Netherlands, which faces an 88 percent higher import tariff. Denatured bioethanol is currently only eligible for mandate compliance in Spain and Sweden. The Dutch legislative change is not forecast to change the trade flows based on the solid EU import demand.

The second origin which could expand exports to the EU is Ukraine. In 2025, Ukrainian bioethanol exports to the EU totaled 110 million liters (100 million liters in 2024). Ukraine has a bioethanol production capacity of about 450 million liters and is anticipated to expand its capacity to about 600 million liters by 2027. Ukraine has an EU TRQ of 125,000 MT (158 million liters) and has been ramping up bioethanol exports to EU over the past decade. Brazil received improved access to the EU with the recent EU – Mercosur deal (see Policy & Programs chapter of this report). The Mercosur deal will be implemented in six phases from May 2026 and covers 450,000 MT (0 percent tariff) ethanol for chemical purposes and 200,000 MT (with a third of the normal import tariff) for other purposes, including fuel. But due to growing domestic demand, expansion of exportable supplies is forecast to be limited. However, for Brazil, as well as for all other suppliers, there are opportunities to export non-crop based advanced bioethanol to the EU, such as bioethanol produced from intermediate crops, low-quality damaged crops, and industrial waste. Overall, FAS Posts projects EU fuel bioethanol imports to increase to 1.91 billion liters in 2026. This forecast is based on a further expansion of bioethanol consumption, while domestic production, despite a record, is not keeping up with this trend.

IV. Biomass-based Diesel

Biomass-based diesel (BBD) includes biodiesel (fatty acid methyl esters, aka FAME) and renewable diesel. Renewable diesel, a full drop-in fuel replacement for fossil diesel, can be produced through various feedstock-technology platform pathways, but the renewable diesel commercialized at scale today is hydrogenation-derived renewable diesel (aka HDRD). HVO (hydrotreated vegetable oil) is an older yet still frequently used term for HDRD. HDRD plants are typically designed to also produce sustainable aviation fuel (SAF). Due to the lack of readily available and accurate supply/demand data on SAF, this report includes SAF in HDRD statistics and therefore BBD statistics as well.

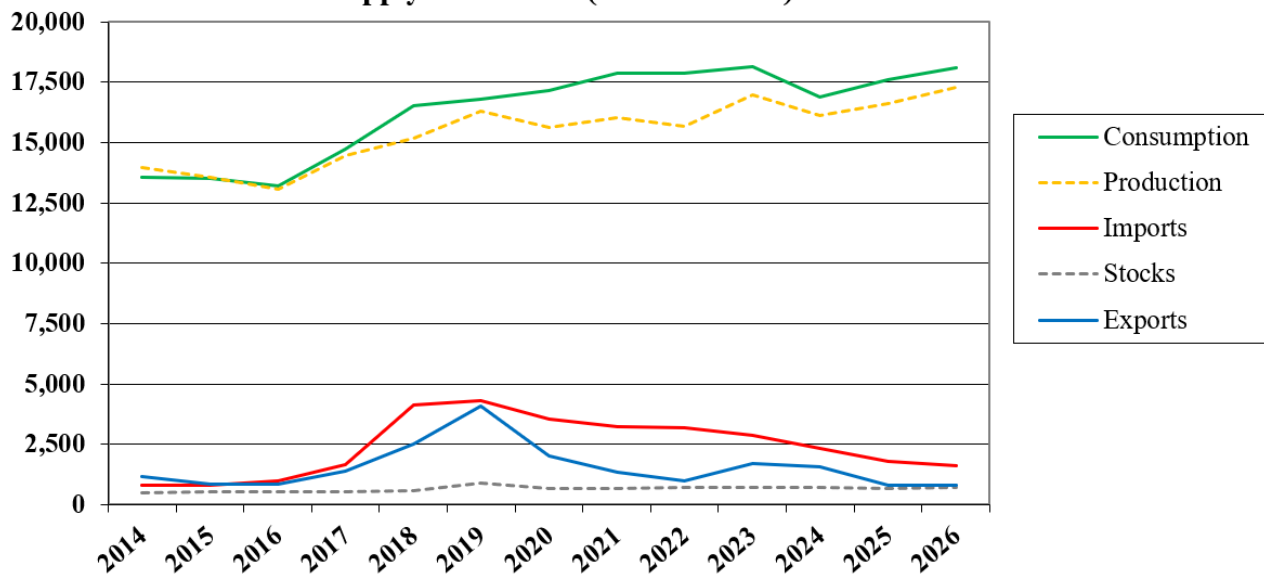
EU Production, Supply and Demand Table

Table 10. FAME, HDRD, and SAF										
(Million Liters)										
Calendar Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026f
Begin. Stocks (excl. SAF)	530	590	900	670	680	715	715	725	675	700
Production	14,464	15,200	16,325	15,629	16,030	15,693	16,995	16,110	16,610	17,308
>HDRD + SAF Production	2,421	2,702	2,842	3,629	4,121	3,482	4,505	4,473	4,759	5,102

Imports (Biodiesel only)	1,669	4,148	4,286	3,545	3,224	3,162	2,870	2,315	1,801	1,600
Exports (Biodiesel only)	1,364	2,530	4,061	2,010	1,346	968	1,695	1,582	796	800
Consumption	14,709	16,508	16,780	17,154	17,873	17,887	18,160	16,893	17,590	18,085
End Stocks (excl. SAF)	590	900	670	680	715	715	725	675	700	723
Production Capacity, Biodiesel (Million Liters)										
Number of Biorefineries	180	179	173	174	171	162	162	136	127	127
Nameplate Capacity	20,109	20,514	19,869	19,824	20,019	20,128	20,362	20,100	20,100	20,100
Capacity Use (%)	59.9	60.9	67.9	60.5	59.5	60.7	61.3	57.9	59.0	60.7
Production Capacity, HDRD (includes SAF) (Million Liters)										
Number of Biorefineries	13	14	15	15	16	17	20	22	22	22
Nameplate Capacity	3,600	3,600	5,372	5,487	6,327	6,437	6,200	7,743	8,281	8,345
Capacity Use (%)	67.3	75.1	52.9	66.1	65.1	54.1	72.7	57.8	57.5	61.1
Feedstock Use for Biodiesel+HDRD+SAF (1,000 MT)										
Rapeseed oil	6,300	6,100	5,950	5,800	6,075	5,800	5,750	5,800	5,950	6,100
UCO	2,600	2,700	3,360	3,500	4,000	3,740	4,000	4,700	4,900	5,100
Palm oil	2,300	2,250	2,600	1,500	890	380	260	80	75	70
Soybean oil	700	1,200	1,070	900	780	950	1,100	850	650	350
Animal fats	860	1,000	1,190	1,250	1,300	1,045	1,000	835	1,000	1,000
Sunflower oil	230	250	270	240	225	300	290	300	270	280
Other	275	500	590	1,400	1,690	2,100	2,500	2,349	2,450	2,625
Biodiesel + HDRD Use in Distillate Pool 1/ (Million Liters)										
Biodiesel+HDRD, On-road	13,450	15,035	15,632	15,831	16,194	15,888	16,299	14,185	14,500	15,000
Biodiesel+HDRD+SAF, Total	14,709	16,508	16,780	17,154	17,873	17,887	18,160	16,893	17,590	18,085
Diesel Pool, On/Off-road /2	231,718	234,226	234,685	207,954	222,068	222,582	217,011	213,581	211,050	205,009
Diesel Pool, Total 3/	311,852	310,425	312,128	288,868	300,840	301,934	290,027	285,116	283,565	277,480
Jet Fuels/Other Kerosene 4/	57,391	60,192	61,361	28,356	34,498	52,325	57,030	59,659	61,778	61,903

Sources/Notes: r = revised / e = estimate / f = forecast EU FAS Posts. 1/ Distillate fuel pool for land transport covering diesel and including all biocomponents (biofuels) and other renewable fuels. 2/ All on/off-road transport; includes construction & agriculture, excludes rail, heavy marine diesels & stationary power. Source: IEA. 3/ Covers all on/off-road uses as defined above plus rail & heavy marine diesels and stationary power. Source: IEA. 4/ Covers all private-commercial-military kerosene-type jet fuels (fossil and bio-based, both Jet A-1 and naphtha-kerosene blend Jet B) + non-aviation applications. Source: IEA. Sources: see chapter VII. Production capacity as of December 31 of year stated. As of 2025 HDRD and SAF trade can be tracked by the HS code 27 10 19 42. The amount of HDRD and SAF included under previous trade codes is unclear. For more information see the Appendix of this report. The feedstock category "other" includes pine oil, tall oil, tall oil pitch, palm fatty acid distillates (PFAD), palm oil mill effluents (POME), empty palm fruit bunches, free fatty acids, and sewage sludge. Beginning/ending stocks: In the absence of reliable data and except for and 2018, data assumes that average stocks equal two weeks of consumption.

Graph 6. Biobased Diesel
EU Supply & Demand (Million Liters)



Source: EU FAS Posts

Consumption

The consumption of BBD¹ is mainly driven by EU Member States' GHG reduction as well as blending mandates and the size of the diesel fuel pool. The latter decreased in 2025 and is estimated to further decrease in 2026 partly because of increasing electrification of the transport sector and the substitution from diesel to gasoline cars. In some cases, subsidies for electrical vehicles or lifting of tax incentives for diesel cars may play a role. Due to GHG reduction mandates, FAME and HDRD with high GHG reduction values have a competitive advantage. Especially in countries that are close to the seven percent volumetric blending limit for FAME (stipulated in the FQD²). Additionally, double counting of biofuels that meet criteria defined by the (revised) REDII limits the effect of increasing mandates on physical blending volumes as less biofuel is needed to fulfill the mandate. The end of double counting advanced biofuels in Germany in 2026, for example is expected to have a significant positive effect on BBD volumes consumed.

In 2025, the consumption of BBD increased by 4.1 percent in comparison to 2024. This is mainly attributed to the first year of SAF mandates, which are expected to continue to drive SAF demand in the future. Higher consumption of BBD is also driven by increases in GHG reduction mandates in different Member States. The effect of increasing GHG reduction mandates is counterbalanced by, for example, significantly cut mandates in Sweden to reduce fuel costs, which led to lower consumption of BBD.

In 2026, the consumption of BBD is expected to increase by 2.8 percent year-on-year. The increase is, similar to 2025, mainly driven by SAF consumption due to the EU's SAF mandate, which increases on a trajectory toward the 2030 mandate. For more information about SAF, see the advanced biofuels chapter of this report. EU jet fuel consumption is expected to remain mostly flat in 2026, as high fuel prices from the Iran conflict increased plane fares and decreased the number of flights.

Increased mandates in most Member States are not expected to affect physical consumption of FAME due to double counting of advanced BBD. Germany is an exception in this regard due to the implementation of revised REDII into national law, which ends double counting retroactively to January 1, 2026. Mandates from 2025 to 2026 did not change in Belgium, Croatia, Estonia, France, Greece, Poland, Romania and Sweden. For more information about the mandates see our FAS GAIN report: [Biofuel Mandates in the EU by Member State – 2025](#), published in January 2026. An update of this report is anticipated to be published in the summer of 2026.

**Table 11. EU BBD¹ Consumption
By Member State (Million Liters)**

Calendar Year	2019	2020	2021	2022	2023	2024 ^r	2025 ^e	2026 ^f
France	3,553	2,997	3,142	3,169	3,124	3,262	3,507	3,501
Germany	2,621	3,583	2,974	2,952	3,038	2,386	3,014	3,232
Spain	2,045	1,810	1,774	1,676	2,307	2,149	2,299	2,350
Italy	1,662	1,654	1,851	1,812	1,904	1,741	1,798	1,798
Poland	1,025	1,076	1,076	1,200	1,259	1,225	1,162	1,170
Sweden	1,744	1,596	1,577	1,781	1,616	746	823	940
Belgium	625	454	738	650	810	810	810	810
Finland	424	381	692	532	532	450	462	474
Austria	578	444	498	472	515	517	520	520
Romania	386	384	453	521	506	501	492	481
Other	2,117	2,775	3,098	3,122	2,549	3,106	2,703	2,809
Total	16,780	17,154	17,873	17,887	18,160	16,893	17,590	18,085

1) Contains small (but increasing) amounts of SAF in most recent years. r = revised / e = estimate / f = forecast EU FAS Posts. Source: FAS EU Posts based on information collected in MT using a conversion rate of 1 MT equals 1,136 liters for FAME and of 1 MT equals 1,282 liters for HDRD. Total consumption may not equal the sum of individual MS production due to rounding error.

Production and Capacity

The EU has a very diverse BBD production structure. FAME production facilities exist in every EU Member State, except for Finland, Luxembourg, Croatia, Cyprus, and Malta. HDRD production is comparatively rare, as it only takes place in about eight EU Member States (see table below). plant size is uniformly larger scale, and the plants are in many cases owned and operated by mineral oil companies. The majority of HDRD capacity consists of dedicated HDRD plants, some of which have been expanded over time to include SAF production. Additionally, co-processing of HDRD with conventional fuel at oil refineries reportedly occurs in Spain, Austria, Portugal, Hungary, Germany, and the Netherlands. It is estimated that total production capacity for BBD increased marginally by less than one percent in 2025 year-on-year. EU BBD production has slightly increased mainly due to higher SAF and HDRD production in anticipation of the new SAF mandate. Overall, EU production capacity remains underutilized for both FAME as well as HDRD and SAF.

In 2025, BBD production increased by slightly over three percent. This is mainly driven by a 6.4 percent increase in HDRD and SAF production due to the new EU SAF mandate. Another factor is the anti-dumping duties on Chinese biodiesel that led to lower imports from China, which created opportunities

for increased domestic production. Imports from China had previously put high pressure on EU BBD production as most certificates for advanced biofuels (RED Annex IX-A feedstock) were imported by the EU. For more details, please see our previous [Biofuels Annual](#). In 2025, the EU lost the United States as their main export market for BBD. Previously, importers in the United States benefitted from a blenders' tax credit, which made importing BBD from the EU, for example, attractive. This changed in January 2025, when the blenders' tax credit changed to a producers' tax credit. Through this change, the market for soybean methyl ester (SME) for export to the United States ended. However, significantly decreased exports to the United States (-100 percent) have led to a higher availability of BBD on the domestic market in the EU. Overall production is forecast to increase by 4.2 percent in 2026.

Table 12. EU FAME Production Main Producers
(Million Liters)

Calendar Year	2019	2020	2021	2022 ^r	2023 ^r	2024 ^r	2025 ^e	2026 ^f
Germany	4,070	3,875	3,937	3,790	4,003	4,086	4,203	4,260
Poland	1,091	1,081	1,138	1,110	1,108	1,150	1,108	1,108
The Netherlands	1,136	1,136	1,136	1,136	1,136	1,136	1,136	1,136
France	2,353	1,851	1,203	1,074	1,031	977	980	980
Spain	1,853	1,698	1,429	1,529	1,150	1,057	1,244	1,250
Belgium	568	568	568	568	704	704	704	704
Italy	616	618	560	551	642	636	642	642
Austria	340	333	335	361	365	335	340	340
Portugal	448	381	363	401	336	336	283	280
Bulgaria	171	137	202	267	314	372	380	380
Czechia	282	294	278	275	296	287	281	290
Other	555	28	760	1,149	1405	561	550	836
Total	13,483	12,000	11,909	12,211	12,490	11,637	11,851	12,206

Ranked by production in 2026, r = revised / e = estimate / f = forecast. Source: FAS EU Posts based on information in MT using a conversion rate of 1 MT equals 1,136 liters. Total production may not equal the sum of individual MS production due to rounding error.

Table 13. EU HDRD¹ Producers
By Member State (Million Liters)

Calendar Year	2019	2020	2021	2022 ^r	2023 ^r	2024 ^r	2025 ^e	2026 ^f
The Netherlands	1,156	1,203	1,247	1,099	1,327	1,167	1,282	1,282
Spain	545	535	409	300	692	944	800	930
Italy	328	797	750	549	814	820	814	820
France	150	476	641	641	641	641	700	700
Finland	424	381	753	610	605	535	603	603
Sweden	208	208	312	278	369	290	449	705
Other	31	29	9	5	57	76	111	62
Total	2,842	3,629	4,121	3,482	4,505	4,473	4,759	5,102

1) Contains small (but increasing) amounts of SAF in more recent years. Ranked by production in 2026; e = estimate / f = forecast. Source: FAS EU Posts based on information in MT, converted to liters using a conversion

rate of 1 MT equals 1,282 liters. Total production may not equal the sum of individual MS production due to rounding error.

Feedstock Use and Co-products Production

Most Member States do not provide official data on feedstock use for the production of FAME, SAF, and HDRD. The figures and analysis below are based on FAS EU Post estimates. In recent years there has been a noticeable shift away from crop-based vegetable oils as feedstocks. Waste and residue-based biofuels are attractive due to their eligibility for double counting in some EU Member States. For further information on the vegetable oil market in the EU, please see the latest FAS GAIN [Oilseeds and Products Annual](#), published April 10, 2026.

Rapeseed oil is historically the main feedstock for BBD in the EU. In 2025, it made up almost 40 percent of total feedstock. In 2026, its total share is expected to remain flat. In general, vegetable oils are only used to produce BBD because the SAF mandate requires the exclusive use of waste and residue as feedstock. Due to higher BBD production in 2026, the quantity of rapeseed oil used to produce BBD is forecast to increase by 2.5 percent. Another driver for the increase in rapeseed oil as feedstock is the end of double counting of advanced biofuels in Germany, which will most likely lead to an increase in demand for rapeseed methyl ester (RME).

Used Cooking Oil (UCO) as a feedstock is very attractive due to its eligibility for double counting in some EU Member States and for its use in SAF production. In both 2025 and 2026 it is estimated to be the second most popular feedstock after rapeseed oil. Its physical quantity is forecast to increase by over four percent year-on-year in 2026 due to multiple reasons, including its eligibility for double counting and rising mandates. Additionally, the EU's new SAF mandate as of 2025 (see Policy and Programs chapter), makes waste and residue feedstock increasingly attractive and SAF production is expected to further increase in 2026. UCO is mostly imported from China, closely followed by Malaysia. Anti-dumping duties imposed on Chinese UCO and BBD may have slowed its growth, but UCO imports from China continue to increase. UCO imports also grew in 2025, mostly driven by the demand for SAF production. Growth in UCO imports is limited by a cap on the amount of UCO that is eligible to be used as feedstock for SAF. Between January and February 2026, UCO imports increased by almost ten percent in comparison to the same timeframe in the previous year. Due to higher demand for SAF, there may be competition for UCO as a feedstock between FAME, HDRD, and SAF production.

Table 14. Import of Used Cooking Oil (UCO)
(HS Code 1518 0095 in 1,000 MT)

	2020	2021	2022	2023	2024	2025	2025 Jan – Feb	2026 Jan - Feb
China	273	619	935	345	478	829	151	161
Malaysia	285	201	204	244	227	175	35	49
United Kingdom	153	150	235	202	140	165	21	48
Russia	99	87	80	84	81	57	10	7
Saudi Arabia	65	67	81	85	97	117	20	15
Argentina	33	24	3	38	51	42	9	7
Other	717	461	442	453	533	660	119	112
Total	1,625	1,609	1,980	1,451	1,607	2,045	365	399

Source: Trade Data Monitor (TDM, sourced from Eurostat), LLC Note: Total imports may not equal the sum of imports by country of origin due to rounding error.

The EU BBD standard DIN EN 14214 limits the volume of inputs categorized as mid-to-high free fatty acid (FFA) content, like *soybean oil*, palm oil, and tallow due to concerns regarding the performance of their resulting FAME in cold weather. On its own, soybean oil methyl ester (SME) does not comply with the iodine value prescribed by this standard (the iodine value functions as a measure for oxidation stability). Since the announcement of the switch from a blending to a production tax credit in the United States in 2024, the amount of soybean oil used as feedstock for biodiesel has decreased yearly. In 2026, the physical amount of soybean oil used as feedstock is forecast to decrease by nearly 50 percent.

The amount of *animal fat* employed as feedstock is difficult to calculate and post's estimates should be viewed as an indication rather than reliable data. The situation varies heavily between different Member States. Spain increased the use of animal fat as feedstock in 2025, particularly for co-processing purposes, to partially replace palm oil mill effluent (POME), given [issues affecting POME's availability](#), which is forecast to continue in 2026.

In 2025, *sunflower oil* is expected to make up two percent of total EU feedstock in 2025 and 2026. Mainly Greece and Bulgaria employ sunflower oil as feedstock for BBD. In 2025, these two Member States alone used over 50 percent of total sunflower oil to produce biofuels, followed by smaller amounts used in Hungary, France, Romania, and Poland. The quantity of sunflower oil used as a feedstock in BBD is expected to increase by almost four percent year-on-year in 2026.

Palm oil is being phased out as a feedstock for biodiesel by 2030 and has, thus, steadily declined since 2020. Palm oil is expected to decline by almost seven percent year-on-year in 2026.

The category “*other*” includes tall oil; free fatty acids; sewage sludge; residues from palm oil production including palm oil mill effluent (POME), palm fatty acid distillates (PFAD), and fresh and empty fruit bunches (FFB and EFB respectively); and cottonseed oil. The share of “other” feedstock in overall feedstock is expected to remain mostly flat in 2026.

Origin of feedstocks and volume of generated byproducts

A large share of EU domestically produced soybean oil is made from imported soybeans. In contrast, most of the rapeseed oil is of domestic origin. The 2026 projection of 6.1 MMT of rapeseed oil used in RME is equivalent to about 15 MMT of rapeseed. This also generates roughly 9 MMT of rapeseed meal as a byproduct, most of which is used for animal feed. Similarly, 2 MMT of soybeans are crushed to generate 0.4 MMT of soybean oil with about 1.4 MMT soybean meal as a byproduct (see also the latest FAS GAIN [Oilseeds and Products Annual](#), published April 10, 2026).

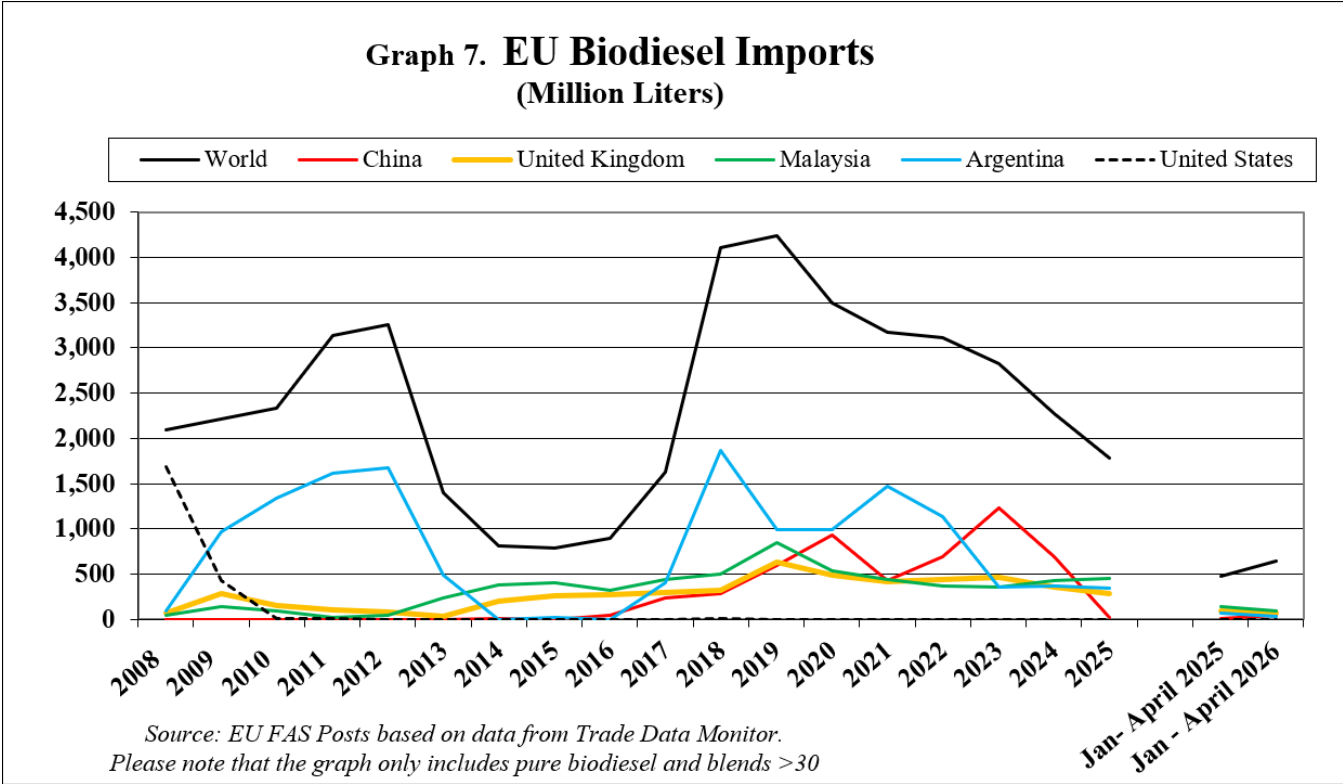
Trade

The trade flows included in this report are limited to biodiesel and petroleum oil containing biodiesel codes CN 38.26.00 and CN 27.10.20 converted to a B100 equivalent. HDRD should be classified elsewhere because it is chemically different from FAME, however, it cannot be ruled out that some quantities are reported under this code. Until the end of 2024, EU Customs combined HDRD with other products under a single 10-digit code that falls under CN 2710.19, and countries known to export HDRD

to the EU and elsewhere also had no HDRD-specific code yet. In January 2025, a new HS-code for HDRD was created (2710.19.42), however, it is unclear if SAF is included in this and what the volume is of non-HDRD products.

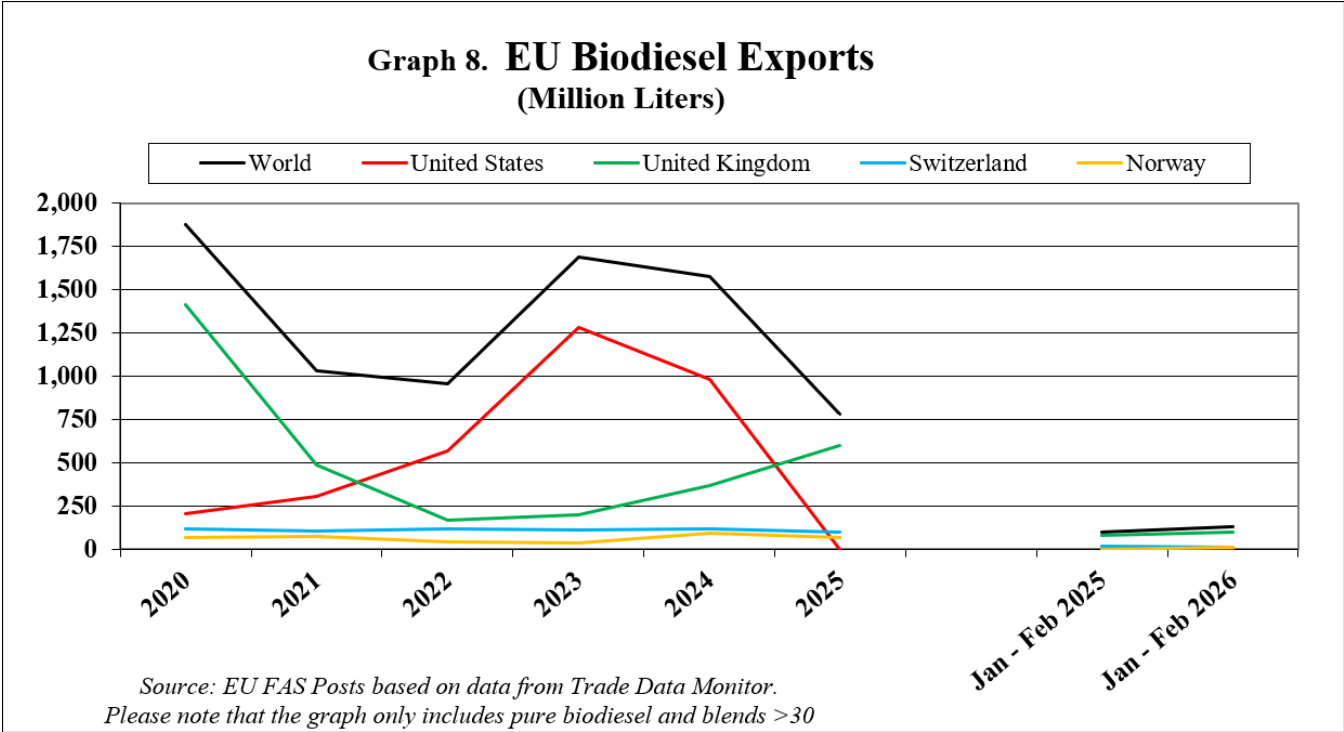
In 2025, the most important suppliers of FAME to the EU were Malaysia, Argentina, and the United Kingdom. Imports from China already decreased significantly year-on-year in 2024 in anticipation of the anti-dumping duties imposed on China by the EU in February 2025, following provisional duties from 2024 (see Policy and Programs chapter). In 2025, the anti-dumping duties led to an almost 100 percent decrease of EU FAME imports from China. This decrease in imports was partially offset by higher imports from Malaysia (+5 percent) as well as South Korea (+81 percent), which is now in fourth place. Overall, imports decreased by almost 22 percent, which opened market opportunities for domestically produced biodiesel in the EU.

In the first two months of 2026, imports from Argentina decreased by 100 percent and imports from Malaysia decreased by 10 percent in comparison to the same timeframe in 2025. FAME imports from Thailand and India increased significantly in the first two months of 2026 in comparison to the same time in 2025. India has historically not been a large supplier of FAME to the EU, while Thailand’s role has increased since 2024. The strongest suppliers to the EU in the first two months were India, Malaysia, South Korea, and Thailand.



Overall EU FAME exports decreased by 50 percent year-on-year in 2025. Until 2025, the United States, United Kingdom, Norway, and Switzerland were the top export destinations for FAME produced in the EU for many years. In 2023 and 2024 especially, most FAME exports from the EU went to the United States. The exports to the United States are likely SME made from U.S. soybeans. The switch from the

blenders’ to the producers’ tax credit effectively ended exports of SME to the United States from the EU. In 2025, FAME exports from the EU to the United States decreased by almost 100 percent. This trend is ongoing in 2026, as exports to the United States remain virtually nonexistent compared to pre-2025 historical volumes. In 2025, the United Kingdom was the largest export market for EU FAME. Overall, FAME exports to the United Kingdom increased by about 60 percent year-on-year. This trend is also reflected in the first two months of 2026 (+25 percent).



Stocks

In the absence of reliable data, estimates and forecasts assume that average stocks are equivalent to two weeks’ supply of consumption.

V. Advanced Biofuels

The EU’s Renewable Energy Directive (RED), extended under REDII and revised on November 20, 2023, establishes an overall policy for the production and promotion of energy using “advanced” (as defined by the EC) biofuels in the EU. Lower-carbon emission biofuels are replacing higher-carbon emission fossil fuels and biofuels (based on full life-cycle analysis) in transportation. EU and national Member State policy is structured to limit further expansion of fossil fuels as well as “conventional” biofuels and incentivize expanded use of “advanced” biofuels. Advanced biofuels (defined as non-food and feed-based biofuels) are less likely to result in land use change because they are made from waste-stream feedstocks or feedstocks that don’t require arable land use. Please refer to the Policy and Programs chapter of this report for more information.

Hydrogenation-derived renewable diesel (HDRD), also known originally as hydrogenated vegetable oil (HVO), is a drop-in fuel that can fully replace fossil diesel. Additionally, with some modification in the production process, plants can co-produce varying levels of sustainable aviation fuel (SAF) or sustainable marine fuel (SMF) along with HDRD. All HDRD, SAF and SMF are treated as an advanced biofuel in this report but would only be considered “advanced” under EU policy when made with qualifying feedstocks including waste-stream feedstocks. The race to electrify transport has narrowed the role biofuels can play and will continue to account for an increasing share of EU and Member State energy use in transportation in the ongoing transition to renewable energy. Hydrogen and so called eFuels (Electrofuels) made with renewable electricity are longer term decarbonization solutions poised to shrink market opportunity for biofuels further, but large market impacts from these fuels are not expected within the coming decade.

Hydrogenation-Derived Renewable Diesel (HDRD)

EU production of HDRD began with Neste Oil in Finland in 2007, then a large expansion followed in 2011 as its Rotterdam plant came online. The feedstocks for HDRD are lipids (plant oils and animals fats, both virgin and waste stream products) and hydrogen. It is a drop-in fuel and fully substitutable for diesel fuel. Pressured to lower the Carbon Intensity (CI) score of HDRD further, EU plants continue efforts to substitute waste-stream feedstock for virgin vegetable oils and are investing in renewable hydrogen production to replace existing fossil-derived hydrogen.

In 2025, HDRD (including SAF and SMF) production increased 6.4 percent to 4.76 billion liters. In 2026, HDRD production is forecast to increase by 7.2 percent to 5.10 billion liters partly based on anticipated expansion of capacity in France and Italy (for more information see Chapter IV Biomass-based Diesel). As a result of projected capacity expansion in the Netherlands, Spain and Portugal, production is forecast to further increase in 2027 and 2028. A major but unknown share of the capacity will be used to produce SAF and SMF. An unknown share of the HDRD is produced by co-processing (production in conjunctions with the processing of fossil inputs). Some of the co-processing plants are listed in Table 15 such as the plants in Spain and Portugal.

The forecast production expansion of HDRD, SAF, and SMF will cause an increase in the demand for feedstocks, in particular the feedstocks listed in Part A and B of Annex IX of the RED (for more information see the Policy and Programs chapter of this report). SAF and SMF produced with food and feed don’t count against the EC targets to decarbonize aviation and marine shipping. The EU increasingly imports feedstocks for advanced biofuel production. In 2025, the EU imported about 2.6 MMT of waste fats and oils under HS1518, 2.0 MMT of palm fatty acid distillate (PFAD) under HS3823, 0.60 MMT of palm oil mill effluent (POME) under HS1522, and 0.25 MMT of tall oil under HS3803.

Advanced Bioethanol

The commercialization of cellulosic bioethanol has lagged far behind the development of HDRD and is essentially not progressing. The main factors that prevent operators from investing in cellulosic bioethanol are high plant construction costs and operational challenges, insufficient regulatory support

(financial incentives), and the challenges of delivering large volumes of bulky, low-energy-content feedstock. Production of bioethanol applying waste streams is however expanding. An increasing number of bioethanol plants process agricultural and food waste products, such as starch sludge and alcohol production residue, which can be counted as advanced biofuels. Production of advanced bioethanol from waste materials listed in Part A of Annex IX of the REDII, is estimated at ten percent of the total bioethanol production (by European members of ePURE for all uses in 2024 estimated by [ePURE](#)).

Sustainable Aviation Fuels (SAF) and Marine Fuels (SMF)

In October 2023, the EU adopted [Regulation 2023/2405](#) on sustainable aviation fuels. The Regulation requires aviation fuel suppliers to ensure all aviation fuel made available to aircraft operators at each EU airport contains an increasing minimum share of SAF. The minimum share of SAF is set at two percent of the total aviation fuel beginning on January 1, 2025. The total EU jet fuel consumption in 2025 is estimated at 61.8 billion liters by the International Energy Agency, which sets the mandated volume at about 1.2 billion liters of SAF. Please refer to the Policy and Programs chapter of this report for more information. In 2024, EU consumption of SAF was 282 million liters, with the Netherlands (147 million liters), France (92 million liters), Sweden (31 million liters), Finland (9 million liters), Italy (2.8 million liters), and Belgium (250,000 liters) as the sole consumers (source Eurostat).

In 2024, the EU production of SAF totaled 178 million liters, with Finland (138 million liters), Sweden (23 million liters), France (15 million liters), Italy (2.7 million liters) and the Netherlands (117,000 liters) as the only producers (source Eurostat). Based on private sector plans, expansion of SAF production will take place throughout the EU, with the main production sites in the Netherlands, Sweden, France, Spain and Italy. The plant capacities are reportedly sufficient to cover the estimated demand of 1.2 billion liters in 2026. For more information see the HDRD section below.

Another opportunity for biofuels is the marine fuel market. In September 2023, the EU adopted [Regulation \(EU\) 2023/1805](#) on the use of renewable and low-carbon fuels in maritime transport (for more information see chapter Policy and Programs). According to the [European Maritime Safety Agency](#), biofuels offer medium and long-term marine fuel alternatives that can enter the market relatively quickly. SMF produced through hydrogenation has the potential to replace conventional fossil maritime fuels without substantial modifications to ship engines and related infrastructure such as storage tanks and fuel pumps.

In 2024, EU consumption of SMF was 308 million liters, with the Netherlands (306 million liters) and France (1.6 million liters) as the sole consumers (source Eurostat). Beginning on January 1, 2026 with the implementation of the revised second Renewable Energy Directive (REDII), the Netherlands included obligations for marine and inland bunker suppliers, which must reduce the GHG emissions of their fuels by respectively 2.9 percent and 2.5 percent in 2026 (all renewable energy included, increasing to respectively 8.2 percent and 14.5 percent in 2030). This Dutch system aims to back the EU Emission Trading System (ETS) and the FuelEU Maritime regulation which are directed toward shipowners to reduce the GHG emissions of the fuel they buy from the bunker suppliers. For more information about the SMF market in the EU and the Netherlands the GAIN Report – [U.S. Biofuels Exports to the Netherlands Surge](#) - New Markets Arise, published May 4, 2026. The volume of bioethanol used by the

maritime sector for transport fuel is limited in the EU, at only 7.4 million liters in 2024 (source Eurostat). The bunkering of bioethanol generally requires more adaptation to the supply chain. But recently a [pilot](#) was successfully conducted in the port of Rotterdam.

Commercial Production of Advanced Biofuels

The table below outlines the operational or close to operational advanced biofuel plants, at a commercial scale, in the EU.

Table 15. Advanced Biofuels Plants in the EU				
Country	Biofuel	Feedstock	Annual Capacity (million liters)	Year of opening
Finland (Porvoo)	HDRD/SAF	Oils and fats	430 (2 lines)	2007
The Netherlands (Delfzijl)	Methanol	Biogas	75	2010
Spain (various locations)	HDRD	Palm oil products and UCO	945 (8 plants)	2011
The Netherlands (Rotterdam)	HDRD	Waste oils and fats	1,280	2011
Italy (Venice)	HDRD/SAF	UCO, animal fats, and residues from the agri-food industry	513 ^a	2014
Finland (Lappeenranta)	HDRD	Tall oil	115	2015
Sweden (Piteå)	HDRD	Tall oil	335	2015
Portugal (Sines)	HDRD	UCO and animal fats	35	2017
France (La Mède)	HDRD/SAF	Oils and fats	640	2019
Italy (Gela)	HDRD/SAF	UCO, animal fats, and residues from the agri-food industry	945	2019
Sweden (Örnsköldsvik)	Methanol	Pulp mill side-streams	6	2020
Austria (Hallein)	Bioethanol	Wood sugar	30	2020
Sweden (Lysekil)	HDRD/SAF	Pyrolysis Oil	900 ^b	2021
Belgium (Ghent)	HDRD	Waste and residues	150	2022
Italy (Taranto)	SAF	UCO, animal fats, and residues from the agri-food industry	12	2022
Spain (Cartagena)	HDRD/SAF	UCO or ag waste	315	2024
Sweden (Gothenburg)	HDRD/SAF	UCO and tall oil	250	2024
Austria (Vienna)	HDRD/SAF	Oils and fats	200	2024
Poland (Gdańsk)	HDRD/SAF	Virgin and waste oils	385	2024
Croatia (Rijeka)	HDRD/SAF	Waste oils and fats	200	2025
France (Grandpuits)	HDRD/SAF	Agricultural and food industry waste	510	2026
Italy (Livorno)	HDRD/SAF	UCO, animal fats, and residues from the agri-food industry	640	2026

Source: EU FAS Posts. Note that the plant numbers and capacities between table 10 and 15 don't match because table 10 is based on the average of the year and table 15 is based on the (forecast) capacity at the end of the year. a. Capacity will be increased to 770 million liters in the years to come. b. Capacity increased to 900 million liters in 2025.

HDRD, SAF, SMF, and Pyrolysis Oil

The Netherlands: In 2011, Neste opened an HDRD plant with an annual capacity of roughly one billion liters in Rotterdam. In addition to HDRD, the Neste plant produces SAF, renewable feedstock (naphtha, propane, and alkanes) to produce polymers and other chemicals. Current annual production capacity at the plant in Rotterdam is a maximum of 1.28 billion liters. In 2024, roughly 90 percent of the feedstock used by Neste to produce HDRD consisted of [waste and residue feedstocks](#). The waste and residues consist of used cooking oil (UCO), by-products of vegetable oil refining such as palm fatty acid distillate (PFAD), tall oil, and oils and fats removed from wastewater. Neste is expanding its refinery in Rotterdam, increasing capacity to roughly 3.45 billion liters (biofuels and intermediate feedstocks), of which roughly 1.5 billion liters is SAF. The company's target is to start up the new production unit in 2027. Shell and BP have announced that they are abandoning their plans to build a biofuels facility in Rotterdam due to insufficient profitability, high construction costs, and deteriorating market conditions.

In 2028 or latest 2029, [SkyNRG](#) is planning to produce SAF for Schiphol Airport in Delfzijl, a seaport in the Northern part of the Netherlands. With the technical expertise of Shell Aviation, the plant will convert waste fats and oils into SAF with a production capacity of 125 million liters. The company reportedly has an off-take agreement to deliver the SAF to KLM for ten years.

Finland: Neste operates a plant with 2 lines each producing roughly 200,000 MT of biofuels and intermediate feedstocks per year (of which about 215 million liters is HDRD). For more information about Neste see under the Netherlands. In 2015, UPM opened an HDRD plant in Lappeenranta. The capacity of the plant is 115 million liters of advanced biofuels per year. Biocrude oil as feedstock for HDRD production is produced by 2 refineries in Finland. [Green Fuel Nordic Oy](#) partnered with a Dutch company, BTG, to produce 25 million liters of pyrolysis oil at its plant in Lieksa. [Fintoil](#) built a crude tall oil refinery with a capacity of 200,000 MT (as feedstock for roughly 100 million liters of HDRD), which became operational in the fall of 2022.

Spain: Spanish HDRD production is mainly from co-processing by petroleum refineries (crude oil refineries co-fed with biogenic lipids). CEPSA (since July 2011), Repsol, (since 2013) and BP (2016) are producing HDRD. During the second quarter of 2024, Repsol started to produce HDRD in Cartagena in an HDRD stand-alone facility. The annual production capacity is estimated at about 315 million liters. Repsol is reportedly planning to open other plants in Puertollano, Coruna, and Tarragona. Cepsa is reportedly planning to open a new plant in Huelva in 2027. In March of 2024, Repsol and Bunge [announced](#) a partnership agreement to develop new opportunities to meet the growing demand for lower carbon intensity feedstocks in renewable fuel production.

Italy: In 2014, Enilive (Eni's company focused on mobility products and services) opened an HDRD plant in Venice, Italy. In the years to come, the capacity is forecast to increase from 510 to 770 million liters per year. Following the model adopted for Venice, Enilive converted their petroleum refinery in Gela, Sicily, into a renewable diesel production facility, with an annual capacity of 945 million liters. In October 2022, Enilive stopped importing palm oil for its Gela and Venice biorefineries ahead of the

deadline set by EU regulations for 2023. Instead of palm oil, the plant processes waste and residue feedstocks (e.g. UCO, animal fats, and residues from the agri-food industry). Enilive plans to increase its biorefining annual capacity to over 6.4 billion liters by 2030, including 2.6 billion liters of SAF. These targets will be supported by ongoing projects at the Venice biorefinery and the construction of new biorefineries in Livorno (Italy), Sannazzaro de'Burgondi (Italy), Priolo (Italy), Malaysia, and South Korea.

Sweden: [VAROPreem](#) sources a diverse range of raw materials, including raw tall oil diesel from SunPine, the world's largest producer of raw tall oil, and food waste such as used cooking oil. The company is planning to further expand its HDRD/SAF production to 5 billion liters in 2035. To achieve this, a [plant of 900 million liters](#) became operational in 2025 in [Lysekil](#). In April 2024, the Finnish company, St1 and SCA [opened](#) a plant to produce up to 250 million liters of HDRD/SAF in Gothenburg, Sweden. St1 is also investigating the construction of another plant with a capacity of 500 million liters to begin operations in roughly 5 years. One of the raw materials which will be used by VAROPreem and St1 for their expanded production is biocrude oil made from tall oil. To increase the supply of biocrude oil, [SunPine](#) increased its capacity from about 100 million liters to 150 million liters in 2021. [Pyrocell](#), owned by VAROPreem and Setra, constructed a plant to produce nearly 30 million liters of biocrude oil from saw dust. In September 2021, production of non-fossil oil started. The pyrolysis oil is refined into renewable diesel and gasoline at VAROPreem's refinery in Lysekil. Both Sweden and Finland increasingly use tall oil as feedstock for HDRD production. For more information see the FAS GAIN Report - [United States Tall Oil Exports to the Nordics Surge](#), published November 6, 2024.

France: Total Energies' HDRD plant located in La Mede (Southern France) began producing HDRD in July 2019. This plant has a maximum capacity of 640 million liters per year. On March 21, 2025, the major French oil company TotalEnergies announced the conversion of the Grandpuits oil refinery near Paris into a zero-crude oil facility, set to be operational by mid-2026. The biorefinery, with a processing capacity of 510 million liters per year, will produce HDRD, SAF, and renewable naphtha, from over 75 percent waste and residues. SAF produced in Grandpuits will be shipped to the two large Paris airports (Charles de Gaulle Airport and Orly Airport) nearby.

Portugal: Portuguese HDRD production is mainly produced through co-processing of vegetable oils by petroleum refineries. Since 2017, [GALP](#) has been producing HDRD in their facilities in Sines. Production capacity is estimated at 35 million liters per year. GALP has announced the installation of a second HDRD unit in Sines, which could have an annual production capacity of over 270 million liters per year in a joint venture with Japan's Mitsui. This project is expected to be operational in 2026. This plant will be capable of switching between HDRD and SAF production using waste fats and oils and residual fats.

Poland: PKN ORLEN confirmed the construction of an installation to produce 385 million liters of HDRD per year in Plock. It will be adapted to process a wide range of lipid raw materials, including rapeseed oil and UCO. The plant will produce both biocomponents for HDRD and SAF. Production of HDRD began in 2024, while the production of SAF started in 2025.

Advanced Bioethanol

Austria: The company [Austrocel](#), a cellulose producer, began construction of an advanced bioethanol plant in early 2020, using residues from its cellulose production as feedstock. The plant became operational by the end of 2020 and delivered its first shipment of advanced biofuels (1.3 million liters) in January 2021. The Austrocel plant has a capacity of 30 million liters per year. The Austrian sugar, starch, and ethanol producer Agrana, uses residuals of its own starch production as feedstock. Currently, about 30 percent of the feedstock is used to produce about 250 million liters of bioethanol is starch sludge.

In *Bulgaria*, ADM opened an advanced bioethanol plant with a capacity of 60 million liters in 2022. The plant has the technical ability to produce conventional, advanced, and industrial chemical bioethanol making use of byproducts from starch production. Bulgaria began producing advanced bioethanol in 2025. There is no official data about production of advanced bioethanol, however, industry sources estimate national supply at around 15,000 MT in 2025. Currently two bioethanol plants have the technical capacity to produce advanced bioethanol from three approved feedstock.

Poland: On March 1, 2022, the Polish oil company, ORLEN Group, announced that they will build an installation to produce advanced bioethanol from non-food products, mainly straw. Its planned annual capacity is 32 million liters. The plant was planned to be operational by the end of 2025. The bioethanol plant will be built together with a biomass (mainly lignin as a byproduct of bioethanol production) fueled combined heat and power (CHP) plant.

Romania: OMV Petrom was planning to build a second-generation bioethanol operation with an annual capacity of 190 million liters based on straw as a feedstock. But the investment was put on hold because the company concluded that the current technological solutions for this type of bioethanol are not viable. The Romanian government launched an aid-scheme of €500 million for advanced biofuels, including €135 million for bioethanol. The call for projects was launched in January 2026.

Biomethanol

Biomethanol can be used as a platform chemical to produce other biochemicals such as lactic acid and formaldehyde. It can also be used as a transport fuel and blended with biofuels, diesel, and gasoline, or used to produce bio-methyl tertiary butyl ether (bio-MTBE) or bio-dimethyl ether (bio-DME). In the Netherlands, the advanced biofuel plant of [Methanex](#) produces biomethanol from biogas. Currently the plant is idled indefinitely. The plant produced about 75 million liters of biomethanol annually. In 2020, [Södra](#) began production of biomethanol at a pulp mill in southeastern Sweden. The plant has an annual capacity of about 6 million liters which is extracted from pulp mill side-streams. Other companies active in the biomethanol sector include [GIDARA Energy](#), which Advanced Methanol Amsterdam (AMA) facility will convert non-recyclable waste into 110 million liters of biomethanol per year. Additionally, the scale-up [Blue Circle Olefins](#) has secured a site in the Port of Rotterdam for a commercial Methanol-to-Olefins (MTO) plant targeted for 2030, which will consume large volumes of renewable biomethanol to manufacture circular ethylene and propylene.

Biomethane

A wide range of plants produce biogas and bio-liquified natural gas (bio-LNG), some of which are used in transport across much of Europe. An example is the bio-LNG plant of [Renewi](#) in the Netherlands. In Germany, [Verbio](#) is producing biomethane from straw at its plant at the Schwedt/Oder site. The plant has the capacity to produce approximately 140 GWh of biomethane per year. This is equivalent to 14 million liters of diesel on an energy basis— using approximately 40,000 MT of straw. The biomethane is destined for use in the transport sector as bio-CNG/LNG and will qualify to count against the THG-reduction sub-mandate for advanced biofuels under the national implementation of REDII in Germany.

VI. Notes on Statistical Data

Bioethanol

Production: Historical fuel bioethanol production capacity, production, and consumption figures are based on statistics of Eurostat and the [European Renewable Ethanol Association \(ePURE\)](#). Production of fuel bioethanol: Eurostat statistics cover indigenous “bioethanol” production (ethanol produced from biomass and/or the biodegradable fraction of waste, to be used as biofuel). Missing MS production figures of Eurostat are estimated by FAS Posts. EU current forecasts are FAS Post estimates. FAS Post based estimates on market information from private sector, national industry organizations, and government sources.

Consumption: Eurostat statistics of final consumption (energy use by transport sector) of blended “biogasoline,” which includes bio-Ethyl tert-butyl ether (bio-ETBE), and negligible volumes of bio-Methyl tert-butyl ether (bio-MTBE), and biomethanol (in total with a maximum of about six percent). Bio-ETBE is not included in the reported bioethanol production but is included in the consumption and trade figures. The volume of bio-ETBE is converted to bioethanol, which contains 45 percent bioethanol by weight. EU current forecasts are FAS Post estimates. FAS Posts based their estimates on market information from private sector, national industry organizations, and government sources.

Trade: Fuel bioethanol import figures are based on Trade Data Monitor (TDM) data (sourced from Eurostat). As the EU has no Harmonized System (HS) code for bioethanol used as fuel, actual trade in bioethanol used as fuel vs other industrial chemical applications are difficult to assess. From 2017 to 2025, the estimation of the EU fuel bioethanol import figures is based on EU imports through preferential trade under HS 2207 (Bolivia, Costa Rica, Guatemala, and Peru), and the imports from Brazil, Canada, and the United States. The monthly shipments larger than 1 million liters plus a unit value of less than \$1.00 per liter are counted as fuel bioethanol, the remainder is treated as non-beverage, non-fuel bioethanol. In addition, bioethanol imports from Ukraine are counted as fuel bioethanol. HS code 29091910 covers ETBE which contains 45 percent ethanol by weight. From 2016 to 2020, EU fuel bioethanol exports are the residual of the production, supply, and demand (PSD) balance. From 2021 to 2026, EU fuel bioethanol exports are anticipated to be minimal at an estimated 100 million liters per year.

Some bioethanol is possibly imported under other HS codes; 220890, 38249091, 38249097, 38249098 and 38249099, blended with petrol or other products. Stocks are the residual of the production, supply, and demand balance.

Feedstock and Co-products: Official data for feedstock use is scarce and generally unavailable from industry and government sources. The figures in this report represent FAS Posts estimates supported by staff assessments of grain and sugar markets which are published by [FAS GAIN reports](#), official EU Member State feedstock use figures of consumed biofuels (not produced), and official trade figures of feedstocks. Feedstock and co-product figures are cross-checked with fuel bioethanol production figures published in the bioethanol balance table using known feedstock yield rates (listed in the Appendix II) to ensure accuracy.

Biodiesel/HDRD

Production and Consumption: Original biodiesel/HDRD data is collected in MT, then converted to liters using a conversion rate of 1 MT = 1,136 liters for biodiesel; 1,282 liters for HDRD. 2017-2022 figures are based on Eurostat and MS official statistics and adjusted by EU FAS Posts using additional market information obtained from national industry organizations and government sources. 2023-2026 figures are FAS post estimates/forecasts.

Trade: Figures are based on Trade Data Monitor (TDM) data (sourced from Eurostat) and the U.S. Bureau of Census and adjusted for U.S. exports of biodiesel blends. A specific customs code for pure biodiesel (B100) and biodiesel blends down to B96.5 (HS 3824.90.91) was first introduced in the EU in January 2008. In January 2012, the code was changed to HS 3826.00.10 for blends containing at least 96.5 percent biodiesel, HS code 3826.00.90 (containing between 30 and 96 percent biodiesel), and HS 2710.20.11 for blends containing at most 30 percent biodiesel. In this report it is assumed that these codes represent a blend of 99, 95, and 5 percent, respectively. Either exports or imports are used as a residual to balance the table and may not exactly match totals calculated using the above methodology.

Feedstock: Official data for feedstock use is scarce and generally unavailable from industry and government sources. The figures in this report represent FAS Posts estimates supported by staff assessments of oilseed markets which are published. Feedstock figures are cross-checked with biodiesel/HDRD production figures as published in the biodiesel/HDRD balance table using known feedstock: biofuel yield rates (listed in the Appendix II) to ensure accuracy.

Fuel Pools: Source for all fuel pools: IEA, Oil Market Report, June 2026.

Appendix I – Abbreviations

BBD = Biomass-based diesel (BBD) includes biodiesel (fatty acid methyl esters, aka FAME) and renewable diesel. Renewable diesel, a full drop-in fuel replacement for fossil diesel, can be produced through various feedstock-technology platform pathways, but the renewable diesel commercialized at scale today is hydrogenation-derived renewable diesel (aka HDRD).

Biodiesel = Fatty acid methyl ester (FAME) produced from agricultural feedstock (plant oils, animal fat, recycled cooking oils) used as transport fuel to substitute for petroleum diesel

Bioethanol = Ethanol produced by fermenting the carbohydrate components of plant materials.

BtL = Biomass to Liquid

Bxxx = Blend of mineral diesel and biodiesel with the number indicating the percentage of biodiesel in the blend, e.g. B100 equals 100% biodiesel, while B5 equals 5% biodiesel and 95% conventional diesel.

CEN = European Committee for Standardization (Comité Européen de Normalisation)

DDG = distillers dried grains

EBB = European Biodiesel Board

EC = European Commission

EU = European Union. “EU” in this report refers to EU27.

Exxx = Blend of mineral gasoline and bioethanol with the number indicating the percentage of bioethanol in the blend, e.g. E10 equals 10% bioethanol and 90% conventional gasoline.

FAME = fatty acid methyl ester

FFA = Free fatty acids

FQD = Fuel Quality Directive

GHG = greenhouse gas

GJ = Gigajoule = 1,000,000,000 Joule or 1 million KJ

Ha = Hectares, 1 hectare = 2.471 acres

HDRD = hydrogenation derived renewable diesel (also known originally as hydrotreated or hydrogenated vegetable oil or HVO)

HS = Harmonized System of tariff codes

KTOE = 1000 MT of oil equivalent = 41,868 GJ = 11.63 GWh

MJ = Megajoule

MMT = Million metric tons

MS = Member State(s) of the EU

MT = Metric ton (1,000 kg)

Mtoe = Million tons of oil equivalent

MW = Mega Watt = 1,000 Kilo Watt (KW)

MWh = Mega Watt hours= 1,000 Kilo Watt hours (KWh)

MY = Marketing Year

Nordics = Denmark, Sweden, Finland, Norway and Iceland

PFAD = Palm fatty acid distillate

PME = Palm oil-based methyl ester biodiesel

POME = Palm Oil Mill Effluent

RED = EU Renewable Energy Directive 2009/28

REDII = EU Renewable Energy Directive 2018/2001

Revised REDII = RED II as revised by EU Directive 2023/2413

RME = Rapeseed oil methyl ester

SAF = Sustainable aviation fuel

SME = Soybean oil methyl ester

TME = Tallow methyl ester, biodiesel made from animal fat

Toe = Tons of oil equivalent = 41,868 MJ = 11.63 MWh

UCO = Used cooking oil/recycled vegetable oil

UCOME = UCO-based methyl ester biodiesel

UDB = Union Database for Biofuels
US\$ = U.S. Dollar

Appendix II - Energy Content and Conversion Rates

1 MT ETBE = 0.45 MT of bioethanol
1 MT Gasoline = 1,342 Liters = 1.03 toe
1 MT BtL = 1,316 Liters = 0.80 toe
1 MT of HDRD = 1,282 Liters = 1.00 toe
1 MT Ethanol = 1,267 Liters = 0.64 toe
1 MT Diesel = 1,195 Liters = 1.02 toe
1 MT Biodiesel = 1,136 Liters = 0.90 toe
1 MT Pure veg Oil = 1,087 Liters = 0.83 toe

Feedstock: Ethanol Conversion Rates

Corn kernels: 1 MT = 402 to 417 liters (has risen since 2006)
Wheat kernels: 1 MT = 393 liters
Rye/Barley/Triticale kernels: 1 MT = 241 liters
Sugar beets: 1 MT = 95 liters

Feedstock: Biodiesel Conversion Rates

Rapeseed oil: 1 MT = 1,136 liters
Soybean oil, crude: 1 MT = 1,113 liters
Soybean oil, 1x refined: 1 MT = 1,128 liters
Sunflower oil: 1 MT = 1,136 liters
Other: 1 MT = 1,136 liters
Crude palm oil (CPO): 1 MT = 1,087 liters
Animal fats/grease: 1 MT = 1,043 liters
Used cooking oil (UCO): 1 MT = 1,043 liters

Ethanol: Co-product Yield Rates (maximum theoretical yield)

Corn kernels: 1 MT = 313 kg of DDG + up to 29 kg of corn oil
Other grain kernels: 1 MT = 313 kg of DDG (negligible vegetable oil)

Appendix III - Related Reports from USEU and MS Posts

Country	Title	Date
Netherlands	U.S. Biofuels Exports to the Netherlands Surge	05/04/26
EU	European Union to Phase Out the Use of Soy-based Biofuels	04/27/26
EU	Grain and Feed Annual	04/20/26
EU	Sugar Annual	04/20/26
EU	Oilseeds and Products Annual	04/10/26
Germany	RED III Implementation Hot Topic at Fuels of the Future 2026	03/25/26

EU	Biofuel Mandates in the EU by Member State - 2025	07/16/25
EU	Wood Pellets Annual 2025	07/14/25
EU	Sugar Annual 2025	04/18/25
EU	Grain and Feed Annual 2025	04/16/25
EU	Oilseeds and Products Annual 2025	04/09/25
Netherlands	Overview of the Market of Industrial Hemp in the Netherlands and Its Potential as Bio-based Material	02/03/25
UK	Sustainable Aviation Fuel in the UK	01/14/25
Finland	United States Tall Oil Exports to the Nordics Surge	11/06/24
UK	Wood Pellets Annual 2024	11/04/24
EU	Biofuels Annual 2024	08/13/24
EU	Wood Pellets Annual 2024	07/05/24
Denmark	Markets for Wood Chips in Northwestern Europe	04/23/24

The GAIN Reports can be downloaded from the following FAS website:

<https://www.fas.usda.gov/data>

This report was a group effort of the following FAS analysts:

Ornella Bettini of FAS/Rome covering Italy
 Sophie Bolla of USEU/FAS Brussels
 Mila Boshnakova of FAS/Sofia covering Bulgaria
 Logan Clow of FAS/London covering Ireland
 Tania DeBelder of USEU/FAS Brussels covering Belgium
 Monica Dobrescu of FAS/Bucharest covering Romania
 Dimosthenis Faniadis of FAS/Rome covering Greece
 Bob Flach of FAS/The Hague covering the Netherlands and the Nordics
 Jana Fischer of FAS/Prague covering the Czech Republic and Slovakia
 Anna Galica of FAS/Warsaw covering Poland and the Baltic States
 Gellert Golya of FAS/Budapest covering Hungary
 Marta Guerrero of FAS/Madrid covering Spain and Portugal
 Andreja Misir from FAS Zagreb covering Croatia
 Alexandra Robu of FAS/Paris covering France
 Katarina von Witzke of FAS/Berlin covering Germany, Austria, and Slovenia

The chapters were coordinated by:

Executive Summary by Bob Flach
 Policy and Programs by Sophie Bolla
 Bioethanol by Bob Flach
 Biomass-based Diesel by Katarina von Witzke
 Advanced Biofuels by Bob Flach

Disclaimer:

This report presents the situation and outlook for biofuels in the EU. This report presents the views of the authors and does not reflect the official views of the U.S. Department of Agriculture (USDA). The data are not official USDA data. Official government statistics on biofuels are not available in many instances. This report is based on analytical assessments, not official data.

Attachments:

No Attachments